Labelling where models disagree: label-efficient comparison of classifiers on group fairness

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Abstract

Comparing classifiers on group fairness needs true labels, far more under simple random sampling than for accuracy, because group rates have small denominators and contrasts are differences of group gaps. We study how to spend a fixed labelling budget when several trained models are compared on group-specific true and false positive rates and error, using two facts: models differ only where their predictions differ, which is rare, and group membership is known before labelling. The method, PATTERN-DIF, stratifies by group and by the number of models predicting the positive class, allocates labels by a weighted Neyman rule with the models' probabilities as proxies, and estimates all contrasts with a stratified difference estimator. We give Wald and simultaneous max-t tests of the contrasts with proofs of asymptotic size, a power function whose efficiency ratio equals the design effect, familywise error control, and a power-optimal allocation. Disagreement sampling and model-assisted estimation are not new; we contribute their combination and the testing theory. Over eleven settings with nine models (six with arbitrary groups), the median efficiency relative to simple random sampling is 5.8, 6.3 and 6.9 at 250, 500 and 1000 labels, against 3.7, 3.9 and 4.4 with the plain estimator and 1.0 for group strata alone. Simulated tests have nominal size and the labels needed for equal power match the prediction within 20%. The worst contrast covered 0.59 at 250 labels.

Keywords: algorithmic fairness, model comparison, stratified sampling, hypothesis testing, multiple testing, label-efficient evaluation

1 Introduction

A model deployed in lending, hiring, health care or public administration is rarely judged on accuracy alone. Auditors and regulators also ask whether its errors fall evenly on the groups it affects, and the usual way of putting that question into numbers is to compare group-specific true and false positive rates, as in the equal opportunity and equalised-odds criteria of Hardt et al. [1]. Unlike demographic parity, these criteria depend on the true outcome, so every instance that enters the calculation must be labelled, and labels are often expensive: a clinician may have to review a record, or an investigator to resolve a case. The question also seldom concerns one model. A team typically has several candidate classifiers of similar accuracy, trained with different methods or settings, and wants to know which is fairer, by how much, and whether the difference can be told apart from sampling noise. This paper asks how to answer that with as few labels as possible.

Fairness comparisons need many labels for two reasons. Group rates are ratios with small denominators: a true positive rate in a minority group is estimated from that group's positives only, which may be a few per cent of a test sample. And the

quantity of interest is a difference of differences, the gap between groups for one model compared with the gap for another, so the variance of four group rates enters every contrast. Statistical work on fairness metrics has noted the resulting instability, particularly for small groups [2, 3]. An earlier study of ours of nine models on twelve data settings (the eleven used below plus Nomao, which is excluded later; not part of this paper, results in `results/realdata_validation.json`, `results/realdata_multi.json` and `results/certify_realdata.json`), using held-out pools as a reference population, gave two illustrations. With 500 randomly drawn test labels, the model with the best equal-weight composite of error, calibration, robustness and equalised-odds gap coincided with the reference winner in a median of 36% of test samples (range 12–96% across settings). And when the test sample size needed to certify, on a single criterion, that a chosen model is within 0.01 of the best was extrapolated from pilot certificates, the median requirement was about 1,100 random labels for error but about 135,000 for the equalised-odds gap. A comparison that is cheap for accuracy is two orders of magnitude more expensive for fairness under simple random sampling. Budgets of that size are rarely available, so the labels have to be spent where they are informative.

Two facts about the problem make this possible. First, competing models disagree on few instances. For a pair of models m and j, the difference in true positive rate within group g is a sum over the positives of that group of $\hat{y}_{mi} - \hat{y}_{ji}$, divided by the number of positives P_g; the summand is zero wherever the two models make the same prediction. The same holds for false positive rates and for error. Among the three most accurate of our nine models, the average pairwise disagreement rate is between 3.0% and 9.5% of the pool in ten of our eleven settings and essentially zero in the eleventh. Near-equally accurate models are known to disagree on individual predictions [4]; here the disagreements are numerous enough to matter for fairness but rare enough to be targeted. Second, group membership is known before labelling. Unlike the outcome, the protected attribute is usually recorded with the instance, so the pool can be divided by group in advance and a small group sampled more heavily than its share. Simple random sampling uses neither fact and spends most of its budget on instances where all models agree and on the majority group.

The problem is as follows. We are given a pool of N unlabelled instances, M fixed classifiers with their binary predictions and predicted probabilities on every instance, and the group of every instance. A budget of L labels may be spent once, before any label is seen. We wish to choose which instances to label and how to estimate, for every pair of models, the difference between their group gaps in true positive rate, the difference between their group gaps in false positive rate, and the difference between their error rates, together with confidence intervals that are valid under the sampling design. Inference is conditional on the trained models, and the target is the value of each contrast on the pool, the natural reference when the pool is the population of use or a large sample from it.

Our method, PATTERN-DIF, has two components. The first is a design. The pool is stratified by group and by the number of models that predict the positive class, so that the instances on which all models agree form the two agreement strata of each group and the strata in between collect the instances on which the models split. Labels are allocated across strata by a weighted generalised Neyman rule that uses the models' averaged predicted probabilities as a proxy for the unknown labels, and are drawn by stratified simple random sampling without replacement. The second component is an estimator. Every contrast is a smooth function of a few population totals, and the proxy means of those totals are known for the whole pool before any label is bought. The stratified difference estimator, in the model-assisted tradition of survey sampling, adds the stratified expansion of the labelled residuals to the proxy total and is design-unbiased for any proxy, however poor. Contrasts are obtained by plug-in and variances by linearisation, which gives Wald intervals. The plain expansion estimator with the same design, PATTERN, is the special case without model assistance. The agreement strata are never left empty: the denominators P_g and Q_g of the group rates involve every instance of the group, including those on which all models agree, so a design that labelled only disagreements would identify the numerators of the contrasts but not the rates themselves.

No ingredient is new on its own, and we state the contribution narrowly. Sampling where models disagree in order to compare their risks goes back at least to Sawade et al. [5], who derived the power-optimal sampling distribution for the difference in risk of two models, noted that it favours instances of disagreement, and extended it heuristically to several models; related ideas appear in active model selection [6, 7] and in discordant-pair evaluation [8]. In biostatistics, comparing two screening tests by verifying discordant or test-positive subjects is decades old and covers sensitivity and specificity, as ratios when the doubly negative cell is not verified [9–11], and optimal two-phase allocation for these quantities has been worked out [12]. Stratification on classifier scores with optimal allocation is established for estimating the performance of one classifier [13–15]; the use of predictions on a whole pool to sharpen an estimator from a small labelled sample is the idea of prediction-powered inference [15, 16] and, much earlier, of model-assisted survey estimation [17, 18]; and label-efficient estimation of group true and false positive rates for one model, or for several models with a given labelled set, has been studied with Bayesian, semi-supervised and mixture-model methods [19–21]. We add the combination of these ingredients, adapted to ratio-type group-fairness contrasts between several models under a fixed labelling budget, with theory and empirical evidence on its performance. Specifically, we (i) define a stratification that crosses the protected group with the models' agreement pattern and an allocation that minimises, up to finite-population corrections and integer rounding, a weighted sum of the linearised variances of all pairwise contrasts at once; (ii) give design-based theory for the plain and the difference estimators, including the way the linearised variance of a ratio contrast draws on disagreement strata through its numerator and on all strata of a group through its denominator, the condition under which the difference estimator beats the plain one (a within-stratum slope of the influence value on its proxy above one half), the design effect, the allocation, and the effect of a misspecified proxy; (iii) give Wald, non-inferiority and simultaneous max-t tests of the contrasts, with proofs of asymptotic size, a power function whose efficiency ratio equals the design effect (so that the labels needed for a given power shrink by that factor), strong familywise error control over all contrasts, and the allocation that maximises the worst-case power; and (iv) evaluate the method on eleven data settings with nine models, in simulations with a known truth, and against baselines and ablations that isolate where the gain comes from. We do not claim novelty for the use of disagreement, for proxy-based Neyman allocation, or for model-assisted estimation with model predictions, and this is not the first label-efficient method for fairness estimation or for comparing several models.

Most of the empirical gain comes from the design. Efficiency is the ratio of the mean squared error under simple random sampling to that under the method, that is, the factor by which simple random sampling would need more labels; within a setting and budget we take the median over the contrasts that the pool does not determine identically, then the median over settings. For PATTERN-DIF it is 5.8, 6.3, 6.9 and 15.5 at budgets of 250, 500, 1000 and 2000 labels (eleven, ten, eight and six settings; 5.5, 5.8, 6.0 and 9.4 without law school, where the leading models agree on almost every instance), against 3.7, 3.9, 4.4 and 7.3 for the plain estimator on the same design, 1.8 to 2.0 for the difference estimator on a simple random sample, and 1.0 to 1.3 for stratification by group alone with the plain estimator. Stratified prediction-powered inference [15] on score strata is competitive with plain PATTERN (higher efficiency in 18 of 35 cells) but PATTERN-DIF beats it in all 35; the importance-sampling design of Sawade et al. [5] and other alternatives are compared in Section 8. Efficiency is larger when the models disagree less (Spearman correlation -0.58 at $L = 250$ without law school). Coverage of nominal 95% intervals is near nominal on average (0.943 at $L = 250$ and 0.949 at $L = 2000$), but not for the worst contrast: 0.59 for law school at $L = 250$ and 0.82 or more in every other setting at that budget. The evidence has limits: one split and one set of nine models per setting (three further splits for the six largest settings, Section 7.5); six of the eleven settings use arbitrary median splits as groups, which carry no fairness meaning; the gain varies from 3.5 to 12.1 at $L = 250$ outside law school; the variance floor for small strata was designed after seeing the coverage failures, so its benefit is in-sample; and the two-phase result rests on a synthetic population.

The paper is organised as follows. Section 2 reviews related work and positions the contribution. Section 3 fixes the setting, the estimands and the notation. Section 4 describes the stratification, the allocation and the estimators, and Section 5 gives their design-based properties and a numerical check of the theory (Supplementary Section S5 has the details). Section 6 gives the hypothesis tests and their size, power and familywise error properties, and Section 6.7 checks them by simulation. Section 7 reports the study on eleven real data settings, Section 7.5 its repetition over several train–pool splits, and Section 7.6 the inference for the deployment population rather than for the pool. Section 8 compares the design with alternative methods from the literature, and Section 9 isolates the contribution of each component. Section 10 turns the results into practical guidance on when the design pays off and how to choose a budget, and Section 11 discusses limitations and open problems. Proofs, additional theory, baseline details, full ablation tables and data details are in the Supplementary Information.

2 Related work

The problem sits where several literatures meet: the statistical comparison of classifiers, the label-efficient evaluation of machine learning models, the design of studies that compare diagnostic tests, the estimation of fairness metrics, prediction-assisted inference and classical survey sampling. We review each in turn, stating what the work does and how our setting differs.

2.1 Statistical evaluation and comparison of classifiers

The comparison of two classifiers on a common test set is usually treated as a paired problem. McNemar's test [22] uses only the discordant pairs of a two-by-two table of correct and incorrect classifications, and Dietterich [23] studied its behaviour, together with resampled and cross-validated t tests, as a tool for comparing learning algorithms. For areas under correlated ROC curves the nonparametric approach of DeLong et al. [24] is standard. Demšar [25] recommended rank-based tests for comparing several classifiers over several data sets. These methods assume that the test set is fully labelled, so which instances to label does not arise, and that the metric is overall accuracy or a ranking measure rather than a difference between models in group-specific rates. McNemar's use of discordant pairs is nonetheless the simplest instance of the first fact on which our design rests: in a paired comparison of accuracy, concordant instances carry no information about the difference.

Group-fairness criteria that depend on the outcome, such as equal opportunity and equalised odds [1], add group-specific denominators to every comparison, and it is known that several such criteria cannot in general be satisfied at once when base rates differ between groups [26]. We do not take a position on which criterion is appropriate. We take the true and false positive rate gaps and error as given and ask how to compare several models on them with few labels.

2.2 Active testing and active evaluation of one model

A large literature reduces the labelling cost of estimating the performance of a single trained model. Active testing [27] selects test points sequentially with acquisition functions tailored to evaluation and corrects the resulting bias with an importance-weighted estimator; active surrogate estimators [28] instead interpolate the loss with an actively trained surrogate, and the approach has been scaled to large language models with cheap in-context surrogates [29]. Prediction-powered active testing [30] combines the importance-weighted estimator with a prediction-powered control variate, derives variance-minimising acquisition rules and proves asymptotic normality. Bayesian assessment of black-box classifiers selects instances by Thompson-type rules [31]. Importance sampling designs for F-measures and other ratio metrics were derived by Sawade et al. [32] and, for rare categories and extreme class imbalance, by Poms et al. [33] and Marchant and Rubinstein [34]; Yilmaz et al. [35] compared importance and Poisson sampling designs for accuracy and F1 with confidence intervals.

Closer to our design are the stratified approaches. Bennett and Carvalho [13] stratified on classifier scores to estimate the precision of deployed classifiers with an online approximation to optimal allocation, Katariya et al. [36] adaptively refined a stratification for accuracy

estimation, and Kumar and Raj [14] studied stratification on classifier output for risk estimation and reported large variance reductions against simple random sampling. More recently, approximate Neyman allocation from surrogate signals has been used to stratify evaluation pools for generative models [37], and stratified importance sampling with one-shot batch acquisition has been proposed for monitoring deployed classifiers [38]. All of this work estimates the performance of one model, usually a mean or an overall ratio such as precision, and builds strata from that model's scores. Our strata come from the agreement pattern of several models crossed with the protected group, our targets are between-model contrasts in group-specific rates, and the allocation must serve many such contrasts at once. The design is also non-sequential: the whole budget is allocated before any label is seen, which suits labelling campaigns run as a single batch.

2.3 Label-efficient model selection and comparison

When several models are available, labels can be chosen to discriminate between them rather than to estimate each one well. Sawade et al. [5] is the closest precedent for our use of disagreement. They compare the risks of two fixed models under a labelling budget, derive the instrumental distribution that maximises the power of a test of the risk difference, observe that it favours instances on which the models disagree, include a baseline that samples only where the models' predictions differ, and extend the approach to more than two models through a heuristic mixture of pairwise distributions, using the models' own predictive distributions as proxies for the unknown labels. Hara et al. [6] develop an adaptive labelling strategy that minimises the variance of estimated differences in test loss for selecting the best of several candidates. Consensus-driven active model selection [7] uses agreement and disagreement among candidate models in a probabilistic model to decide which points to label and updates a posterior over the identity of the best model. Discordant pair analysis [8] evaluates a new classifier against a baseline of known performance by adjudicating only the cases on which they differ, and Kuboyama [39] reports that labelling every row on which at least two candidates disagree settled the choice of the most accurate model in all 108 of its experimental conditions. Other work selects the best of several pretrained models from a stream or a pool with few labels [40, 41], chooses instances for annotation from clusters of output differences between two text generators [42], or studies whether a fixed budget should be spent on more items or on repeated labels when identifying the more accurate of two classifiers [43]. The only work we found that selects among several models on both accuracy and subgroup fairness under label costs is the Bayesian bandit formulation of Kher et al. [44], in which an online policy decides which agent to query and the target is regret.

This literature shows that disagreement is where comparative information lies. It differs from our setting in three respects. Its target is a risk or an accuracy ranking, not a contrast of ratios with group-specific denominators; most of it is adaptive or Bayesian and aims at identifying a best model, not at estimating contrasts with design-based confidence intervals; and the designs that label disagreements only, such as Musgrove et al. [8] and Kuboyama [39], rely either on known baseline performance or on exhaustive labelling of the disagreement set. For error rates the denominator is the pool size and disagreements suffice, but for true and false positive rates the group prevalences P_g and Q_g depend on the agreement strata too.

2.4 Verification designs for comparing diagnostic tests

Biostatistics has long faced the problem of comparing two screening tests when the reference standard is too costly or invasive to apply to everyone. Schatzkin et al. [9] proposed applying the reference procedure only to people positive on at least one of the two tests and comparing the tests through ratios of sensitivities and of false positive rates; Chock et al. [10] studied the related design in which individuals negative on both tests are not verified; and Pepe and Alonzo [11] showed that, under verification of screen positives only, absolute sensitivities and specificities are not identified but their ratios are, and gave a regression framework for relative accuracy. Correcting for verification that depends on the test result is the subject of a classic literature on verification bias [45], and the practice of resolving only discordant results with

the reference standard, so-called discrepant analysis, has been criticised because it assumes that concordant results are correct [46]. Two-phase designs with a cheap first-phase classification and an optimally allocated second-phase verification were developed for the sensitivity, specificity and predictive value of one test by McNamee [12], within the broader two-phase framework of Neyman [47] and its epidemiological descendants [48, 49].

Our true and false positive rate contrasts between two classifiers are this design problem transplanted to machine learning. The differences lie in scope and in what is identified. The diagnostic literature compares two tests, not several models; it has no protected groups or contrasts of group gaps; its strata are the test results themselves, not results combined with predicted probabilities that set the allocation; and several of its designs leave the doubly negative cell unverified on purpose, so that only relative measures are identified. Our design samples every stratum with known positive probability, so that differences in rates, and not only their ratios, are identified, at the price of spending some labels where all models agree. The allocation rule decides how many such labels are worth spending.

2.5 Fairness estimation and auditing with few labels

Several lines of work estimate fairness quantities with limited data. One line audits a model by querying it rather than by acquiring ground truth: active fairness auditing estimates demographic parity with few queries to the audited model [50], and query-efficient auditing of language models gives intervals for metrics such as group differences in AUC with active selection of queries, reporting large savings over stratified sampling [51]. Demographic parity needs no outcome labels, so the first setting differs from ours; the second paper does address an outcome-dependent metric, for one model. A second line provides valid inference: Chugg et al. [52] give sequential, anytime-valid tests of fairness that allow data to be collected by changing and group-dependent policies, Cherian and Candès [53] construct simultaneous bounds on performance disparities over many subgroups of one fully labelled model, Ferrara et al. [3] adapt hypothesis tests of fairness to group size, and Lum et al. [2] show that common disparity summaries are biased estimators when groups are small and propose corrections. A third line uses unlabelled data to improve estimates of group-specific rates. Ji et al. [19] calibrate scores per group in a hierarchical Bayesian model to infer group differences in accuracy, true positive rate and false positive rate of one model, and remark that stratified sampling by group could help but is not always possible. Gao and Gronsbell [20] impute missing outcomes and augment for robustness to estimate group-specific true and false positive rates and calibration with semi-supervised efficiency gains, Gronsbell et al. [54] evaluate one prediction rule semi-supervisedly when the labels come from a stratified sample, Miller et al. [55] estimate subpopulation performance with model-based metrics, and Das et al. [56] audit fairness under partial feedback with a cost model for acquiring extra labels. Closest in setting is SSME [21], which fits a semi-supervised mixture model to the labels and the scores of several classifiers and estimates any metric for each classifier, including within demographic subgroups.

These methods take the labelled set as given, so there is no design to optimise, or concern one model, or both. Our contribution is complementary: a sampling design and design-based estimators for between-model fairness contrasts. Model-based and semi-supervised estimators could in principle be applied, with design weights, to the labels our design collects.

2.6 Prediction-powered and model-assisted inference

Prediction-powered inference [16] combines a small labelled sample with machine learning predictions on a large unlabelled sample and gives confidence intervals that are valid whatever the quality of the predictions; PPI++ [57] tunes the weight on the predictions so that the interval is never wider than the classical one asymptotically. Stratified prediction-powered inference [15] applies the idea within strata and allocates the label budget across strata by a Neyman-type rule. Active statistical inference [58] samples labels with probabilities driven by model uncertainty and combines the resulting design-unbiased estimator with prediction-powered corrections; robust sampling [59] interpolates between uniform and active

sampling so that a poor proxy does not make matters worse than uniform sampling, and fixed-size variants replace Poisson sampling by conditional Poisson sampling [60]. Prediction-powered ranking [61] gives simultaneous rank-sets for several language models from pairwise comparisons. Cross-prediction-powered inference [62] extends the idea to the case in which the predictor is fitted on the same labelled data by cross-fitting, and semi-supervised mean estimation with a fitted regression is studied by Zhang et al. [63]. In our setting the proxy is a fixed function of the outputs of models that were trained elsewhere and is never refitted on the labelled units, so no cross-fitting is needed and the difference estimator is exactly design-unbiased; the price is that the proxy is not tuned to the labels. In these methods predictions enter either as a control variate in the estimator or as a guide to inclusion probabilities, and the target is typically a single mean or a vector of means for one model. We use predictions in both ways: to set the allocation, as in the stratified and active designs, and in the estimator, through the stratified difference estimator, the finite-pool form of prediction-powered inference with the coefficient on the predictions fixed at one. The targets are vectors of nonlinear contrasts between models, the strata are defined by the agreement pattern and the group, and the pool-wide predictions are the models' own probabilities. Stratified prediction-powered inference is the nearest published estimator; we use it, on strata built from the proxy score, as a baseline in Section 8.

The same idea is much older in survey sampling, where auxiliary information is used to build the design and to sharpen the estimator. Model-assisted estimation, in which a working model improves efficiency while validity rests on the design [17, 18], and calibration estimators [64] are the classical forms, and prediction-powered inference can be read as a member of this family, as can our difference estimator. The novelty lies in the strata, the allocation and the target, not in the estimation principle.

2.7 Survey-sampling foundations

The design is assembled from standard survey-sampling components. Stratified sampling with allocation proportional to stratum size times stratum standard deviation, which minimises the variance of an estimated total for a fixed sample size, is due to Neyman [65]. When several quantities are estimated from one sample, a single allocation must compromise between them; optimum allocation in multivariate surveys has a long history, including analytical solutions [66], and the weighted generalised Neyman rule we use minimises a weighted sum of variances, the simplest such compromise. Unbiased estimation of totals under unequal inclusion probabilities goes back to Horvitz and Thompson [67]. Group rates are ratios of totals, and the standard treatment of ratio estimators and of smooth functions of totals by Taylor linearisation is given by Woodruff [68], Binder [69], Särndal et al. [17] and Wolter [70], with a practical account in Lumley [71]. Two-phase sampling, in which a cheap first-phase measurement guides the selection of a second-phase sample, was introduced by Neyman [47]; our setting is a two-phase design in which the first phase, the models' predictions on the whole pool, costs nothing. These foundations do not supply the choice of strata and of variance targets for comparing classifiers on fairness; that is the part we work out.

2.8 Positioning

Disagreement-based sampling for comparing models, discordant-result designs for comparing diagnostic tests, proxy-based optimal stratification for evaluating one classifier, model-assisted and prediction-powered estimation, and label-efficient fairness estimation for one model are all established, and we build on each. To our knowledge they have not been combined into a fixed-budget design and estimator for fairness contrasts between several classifiers. Such a method must handle group-prevalence denominators that depend on all strata, allocate labels for many contrasts at once, and give design-based confidence intervals. In our evidence, stratification on disagreement contributes more than the model-assisted estimator, and stratified prediction-powered inference on score strata competes with the plain design but not with the combination. The paper presents that combination, its design-based theory, and an assessment of when it pays off and where it does not.

3 Problem setting

3.1 Pool, models and labels

We consider a pool $U = \{1, \dots, N\}$ of instances drawn from the deployment population and $M \geq 2$ binary classifiers that have already been trained elsewhere. For every instance $i \in U$ the following quantities are available before any label is collected: the predictions $\hat{y}_{mi} \in \{0, 1\}$ and predicted probabilities $\hat{p}_{mi} \in [0, 1]$ of every model $m = 1, \dots, M$, and the value $G_i \in \{0, 1\}$ of a binary group attribute. The label $Y_i \in \{0, 1\}$ is unknown. It can be obtained for any instance, at a cost of one unit, and the analyst can afford $L \ll N$ labels. The group attribute is assumed to be recorded for all instances, as is usual when a protected attribute is part of the data used for auditing even if it is not used for prediction. The labels are treated as fixed, and all randomness comes from the choice of which instances to label. Inference therefore refers to the finite pool, the natural target when the pool is the population of use, and extends to the superpopulation in the usual way when the pool is a random sample from it [17].

The classical evaluation protocol starts from a labelled test set of fixed size and asks only how to analyse it; here we also ask how to spend the labelling budget. The setting differs from active learning too: the models are fixed, and labels are collected only to compare them.

3.2 Totals and estimands

All quantities of interest are smooth functions of a small number of population totals, each of which is linear in the labels. For a group $g \in \{0, 1\}$ and model m define

$$P_g = \sum_{i:G_i=g} Y_i, \qquad Q_g = \sum_{i:G_i=g} (1 - Y_i), \tag{1}$$

$$\begin{aligned} TP_{mg} &= \sum_{i:G_i=g} Y_i \, \hat{y}_{mi}, \\ FP_{mg} &= \sum_{i:G_i=g} (1 - Y_i) \, \hat{y}_{mi}, \\ E_m &= \sum_{i \in U} |\hat{y}_{mi} - Y_i|. \end{aligned} \tag{2}$$

Each total has the form $T_c = \sum_{i \in U} t_{ic}(Y_i)$ with $t_{ic}(Y_i) = Y_i u_{ic} + (1 - Y_i) v_{ic}$, where u_{ic} and v_{ic} are known for every instance. For example, TP_{mg} has $u_{ic} = \mathbf{1}\{G_i = g\}\hat{y}_{mi}$ and $v_{ic} = 0$, and E_m has $u_{ic} = 1 - \hat{y}_{mi}$ and $v_{ic} = \hat{y}_{mi}$. We collect the totals in a vector T of length $C = 4M + 4 + M$.

The group-wise true and false positive rates are $\mathrm{TPR}_{mg} = TP_{mg}/P_g$ and $\mathrm{FPR}_{mg} = FP_{mg}/Q_g$. The between-group gaps of model m are

$$\begin{aligned} \Delta_m^{\mathrm{TPR}} &= \mathrm{TPR}_{m1} - \mathrm{TPR}_{m0}, \\ \Delta_m^{\mathrm{FPR}} &= \mathrm{FPR}_{m1} - \mathrm{FPR}_{m0}, \end{aligned} \tag{3}$$

and the equalised-odds gap is $\mathrm{EO}_m = \max(|\Delta_m^{\mathrm{TPR}}|, |\Delta_m^{\mathrm{FPR}}|)$ [1]. A model with small gaps has more similar error rates in the two groups.

Our primary estimands are comparisons between models. For a pair (m, j) let

$$\begin{aligned} \theta_{mj}^{\mathrm{TPR}} &= \Delta_m^{\mathrm{TPR}} - \Delta_j^{\mathrm{TPR}}, \\ \theta_{mj}^{\mathrm{FPR}} &= \Delta_m^{\mathrm{FPR}} - \Delta_j^{\mathrm{FPR}}, \\ \theta_{mj}^{\mathrm{ERR}} &= (E_m - E_j)/N. \end{aligned} \tag{4}$$

A negative $\theta_{mj}^{\mathrm{TPR}}$ means, for instance, that model m has a smaller signed TPR gap than model j. We write $\theta = f(T)$ for the vector of all $3\binom{M}{2}$ comparisons, where f is a known smooth map on the region where $P_g, Q_g > 0$. The primary estimands are the signed gaps, because they are smooth functions of the totals; the non-smooth equalised-odds gap and the ranking of models by it are studied in Section 9.

3.3 Why comparisons on fairness are label-hungry

Under simple random sampling (SRS) of L labels, the TPR of a group is estimated from the positives of that group that happen to be sampled, about $L\,P_g/N$ of them. In the Adult data with sex as the group, female positives make up 3.6% of the pool, so a budget of 1,000 labels yields on average 36 relevant instances for TPR_{m1}. An earlier study of ours on twelve data settings (described in Section 1; not part of this paper) found that certifying a regret of 0.01 in the equalised-odds gap required on the order of 10^5 randomly sampled labels, against about 10^3 for the error rate. Oversampling the minority group alone helps less than

might be expected, because each group-wise rate is a ratio whose denominator must also be estimated, and because the comparison of two models depends on the difference of their predictions, not on each prediction separately.

3.4 Two structural facts

The sampling design can exploit two features of the problem.

Comparisons depend only on disagreements. Write $s_1 = +1$ and $s_0 = -1$. Then

$$\theta^{\mathrm{TPR}}_{mj} = \sum_{g\in\{0,1\}} \frac{s_g}{P_g} \sum_{i:G_i=g} Y_i\,(\hat{y}_{mi} - \hat{y}_{ji}), \qquad (5)$$

and similarly for $\theta^{\mathrm{FPR}}_{mj}$ with $(1-Y_i)$ and Q_g, while $N\theta^{\mathrm{ERR}}_{mj} = \sum_i (\hat{y}_{mi} - \hat{y}_{ji})(1-2Y_i)$. The numerators vanish on every instance where m and j predict the same label. Reasonably accurate competing models agree on most instances; in our eleven data settings the three most accurate of nine models disagree, averaged over their three pairs, on between 0.0% and 9.5% of the pool (Table 2 in Section 7). Labels spent on instances where all models agree inform only the denominators P_g and Q_g.

Everything except the label is known in advance. The group of every instance and the full vector of predictions are known before labelling. A design can therefore condition on them freely, and on the predicted probabilities, without introducing any bias, because the inclusion probabilities are known.

The design of Section 4 uses both: it stratifies the pool by group and by how many models predict the positive class, and allocates labels across strata to minimise the variance of all comparisons at once.

4 Design and estimation

4.1 Stratification

Let $K_i = \sum_{m=1}^{M} \hat{y}_{mi} \in \{0, \dots, M\}$ be the number of models that predict the positive class for instance i. The design, which we call PATTERN, partitions the pool into strata indexed by (G_i, K_i). The two extreme strata of each group, $K_i = 0$ and $K_i = M$, contain the instances on which all models agree; the strata with $0 < K_i < M$ contain every instance on which at least one pair of models disagrees, and the value of K_i indicates how evenly the models are split. Strata with fewer than 50 instances are merged with the adjacent stratum (in K) of the same group, so that within-stratum variances can be estimated; this gives $H \le 2(M+1)$ strata, of size N_h.

The count K_i is a coarse summary of the prediction pattern $(\hat{y}_{1i}, \dots, \hat{y}_{Mi})$, which has up to 2^M values. It is invariant to the ordering of models, keeps the number of strata linear in M, and its extreme values isolate the agreement region exactly. Finer stratifiers are compared in Section 9.

4.2 Estimation

Within each stratum a simple random sample of n_h instances is drawn without replacement and labelled ($\sum_h n_h = L$). The plain stratified expansion estimator of a total T_c is

$$\hat{T}^{\mathrm{HT}}_c = \sum_{h=1}^{H} N_h\,\bar{t}_{hc}, \qquad \bar{t}_{hc} = \frac{1}{n_h}\sum_{i\in s_h} t_{ic}(Y_i), \quad (6)$$

which is unbiased for T_c [72]. It uses the predicted probabilities only to set up the design. The proposed estimator uses them in the estimation as well. Let $\pi_i = M^{-1}\sum_m \hat{p}_{mi}$ be the average predicted probability and $\psi_{ic} = \pi_i u_{ic} + (1-\pi_i)v_{ic}$ the value that $t_{ic}(Y_i)$ would have in expectation if Y_i were Bernoulli(π_i). This quantity is known for every instance of the pool, including the unlabelled ones. The difference estimator corrects the pool-wide proxy total by the labelled residuals,

$$\begin{aligned} \hat{T}_c &= \sum_{i\in U} \psi_{ic} + \sum_{h=1}^{H} N_h\,\overline{(t_c - \psi_c)}_h, \\ \overline{(t_c - \psi_c)}_h &= \frac{1}{n_h}\sum_{i\in s_h} \{t_{ic}(Y_i) - \psi_{ic}\}, \end{aligned} \qquad (7)$$

which is exactly unbiased for T_c for any label-free proxy, good or bad, because the proxy total over the whole pool is known [17, 73]. Setting $\psi \equiv 0$ recovers (6). The comparisons are estimated by the plug-in $\hat{\theta} = f(\hat{T})$, a ratio-type estimator, and its variance by Taylor linearisation [17, Section 5.5]: with the residual influence values

$e_i = \nabla f(\hat{T})^\top \{t_i(Y_i) - \psi_i\}$,

$$\hat{V}(\hat{\theta}_k) = \sum_{h=1}^{H} N_h^2 \Big(1 - \frac{n_h}{N_h}\Big) \frac{s^2_{e_k,h}}{n_h}, \qquad (8)$$

where $s^2_{e_k,h}$ is the sample variance of the kth residual influence values in stratum h. Wald intervals $\hat{\theta}_k \pm 1.96\,\hat{V}(\hat{\theta}_k)^{1/2}$ are reported. If a sampled denominator total is zero (no sampled positive in a group), the estimate is undefined; this happens with small probability under SRS at small budgets and never occurred under the stratified designs in our main study (Section 7).

We call the design with the plain estimator PATTERN and the design with the difference estimator PATTERN-DIF; PATTERN-DIF is the proposed method. The difference estimator removes the part of the label variance that the predicted probabilities explain. The proxy is informative (its area under the ROC curve on the pools is between 0.76 and 1.00) and only moderately miscalibrated (10-bin calibration error between 0.02 and 0.07 in ten of the eleven settings and 0.20 in EEG; the worst probability quintile is off by 0.04 to 0.20; `results/proxy_calibration.json`), and the correction reduces the variance by a further factor of 1.3 to 1.7 in our experiments (Section 7). The theory in Section 5 shows when the correction helps (the within-stratum regression slope of the true influence on its proxy must exceed one half) and when it does not. The weight is fixed at one throughout. Estimating a weight per stratum and estimand, as in prediction-powered inference with tuned weights, added about 2 to 3% in median efficiency and lowered coverage (Sections 8 and 9).

4.3 Allocation

The allocation $(n_1, \ldots, n_H)$ determines the efficiency of the design. For a single estimand the variance of the linearised estimator is minimised by Neyman allocation, $n_h \propto N_h S_h$, where S_h is the within-stratum standard deviation of the influence values. With many estimands we minimise the weighted sum $\sum_k w_k V_k$ of their linearised variances, which leads (Section 5) to

$$n_h \;\propto\; N_h \Big(\sum_k w_k S^2_{hk} \Big)^{1/2}. \qquad (9)$$

Algorithm 1 PATTERN-DIF for comparing M classifiers on group-fairness contrasts

Require: pool U with predictions $\hat{y}_{mi}$, probabilities $\hat{p}_{mi}$, groups G_i; budget L

1: $K_i \leftarrow \sum_m \hat{y}_{mi}$; strata $h \leftarrow (G_i, K_i)$; merge strata with $N_h < 50$ within group
2: $\pi_i \leftarrow M^{-1} \sum_m \hat{p}_{mi}$; proxy totals $\tilde{T} \leftarrow \sum_i \{\pi_i u_i + (1 - \pi_i) v_i\}$
3: $A_i \leftarrow \nabla f(\tilde{T})^\top u_i$, $B_i \leftarrow \nabla f(\tilde{T})^\top v_i$; proxy moments of z_i
4: S^2_{hk}, $S^2_k \leftarrow$ proxy within-stratum and overall variances
5: $n_h \leftarrow$ allocation (9) with $w_k = 1/S^2_k$, minimum 10, capped at N_h, $\sum_h n_h = L$
6: draw n_h instances without replacement in each stratum and query their labels
7: $\hat{T} \leftarrow$ difference estimator (7); $\hat{\theta} \leftarrow f(\hat{T})$; $\hat{V} \leftarrow$ (8); Wald intervals
8: **return** $\hat{\theta}$, standard errors, intervals

We use $w_k = 1/S^2_k$, where S^2_k is the population variance of the influence values of estimand k; this weights each estimand by the inverse of its variance under SRS, so that every comparison counts equally in relative terms.

The stratum variances depend on the unknown labels. We replace them by their expectations under a proxy model in which Y_i is Bernoulli with probability $\pi_i = M^{-1} \sum_m \hat{p}_{mi}$, the average predicted probability of the models. Because the influence values are linear in Y_i, $z_{ik} = Y_i A_{ik} + (1 - Y_i) B_{ik}$ with known A_{ik}, B_{ik} evaluated at the proxy totals, and the proxy moments $\mathrm{E}_\pi z_{ik} = \pi_i A_{ik} + (1-\pi_i) B_{ik}$ and $\mathrm{E}_\pi z^2_{ik} = \pi_i A^2_{ik} + (1-\pi_i) B^2_{ik}$ give closed-form proxy variances. No label is used. A poor proxy can only reduce efficiency; both estimators remain unbiased for the totals because the allocation does not depend on the labels. We use this allocation, computed from the variance of the plain influence values, for both estimators. An allocation from the variance of the residual influence values is theoretically optimal for the difference estimator but gave no gain in our experiments (Section 7), so we keep the simpler rule. Each stratum receives at least ten labels (or all of its instances if it is smaller), and rounding to integers is done greedily so that the budget is met exactly.

4.4 Reference designs

We compare PATTERN-DIF with a ladder of simpler designs, each removing one ingredient, so that the contribution of each is visible. SRS samples L instances without replacement from the whole pool and uses the plain estimator (6). SRS-DIF uses the same simple random samples with the difference estimator (7), which is prediction-powered inference with the proxy as predictor [16]. GROUP stratifies by the group attribute only and allocates with (9); it oversamples the minority group when that reduces variance, the natural first idea for fairness auditing. GROUP-DIF adds the difference estimator. PATTERN is the design of Section 4 with the plain estimator. In Section 9 we also use an allocation computed from the true labels instead of the proxy; it cannot be used in practice and is not an upper bound on efficiency, since (9) minimises a weighted sum of first-order variances rather than the mean squared error of each estimand. Further baselines from the active-testing and survey-sampling literatures, including a stratified prediction-powered estimator, are compared in Section 8.

4.5 Computational cost

The design needs one pass over the pool to compute K_i, π_i and the proxy moments, which costs $O(NMC)$ operations with $C = O(M)$ totals, and the estimation costs $O(LMC)$. For the pools used here ($N \leq 19{,}537$, $M = 9$) in our implementation the design computation took at most 0.6 s per pool and the sampling and estimation of one replicate at most 0.05 s (median times in milliseconds are in the last column of Table 8).

5 Theory

We work in the design-based framework of survey sampling [17, 72]: the pool U, the predictions $\hat{y}_{mi}$, the probabilities $\hat{p}_{mi}$, the groups G_i and the labels Y_i are fixed numbers, and the only randomness is the selection of the labelled sample. The strata, the allocation and the proxy are computed from $(\hat{y}_{mi}, \hat{p}_{mi}, G_i)_{i \in U}$ alone, so they are fixed before any label is seen, and no model for Y given the predictions is needed for validity. Both estimators are unbiased for every proxy; a wrong proxy affects efficiency only. Additional results (closed-form design effects, misspecification bounds, tuning of the difference coefficient) are in Supplementary Section S2, and full proofs in Supplementary Section S1.

5.1 Set-up

Write $t_i(Y_i) = Y_i u_i + (1 - Y_i) v_i \in \mathbb{R}^C$, with entries of u_i, v_i in $\{0, 1\}$, for the contribution of instance i to the totals $T = \sum_{i \in U} t_i(Y_i)$, and $\tau = T/N$. Every comparison estimand is a smooth function of the totals, $\theta = f(T) = \phi(T/N)$, with ϕ infinitely differentiable where the denominators $P_g/N, Q_g/N$ are positive. The *influence value* of instance i is

$$z_i = \nabla f(T)^\top t_i(Y_i) = N^{-1} \nabla \phi(\tau)^\top t_i(Y_i). \quad (10)$$

Strata $h = 1, \ldots, H$ partition U into sets U_h of sizes N_h, $W_h = N_h/N$; n_h labels are drawn from U_h by simple random sampling without replacement, independently across strata. For a variable x let $\bar{x}_h$ and $S^2_{x,h}$ be the stratum mean and variance, $\bar{x}_{s_h}$, $s^2_{x,h}$ their sample versions, $f_h = n_h/N_h$ and $c_h = N_h^2(1 - f_h)/n_h$. The plain (Horvitz–Thompson, Horvitz and Thompson 67) estimator, its linearisation variance estimator [17, Ch. 5] and the linearised variance are

$$\begin{aligned} \hat{T} &= \sum_{h=1}^{H} N_h \bar{t}_{s_h}, \qquad \hat{\theta} = f(\hat{T}), \\ \hat{V} &= \sum_{h=1}^{H} N_h^2 \, \frac{1 - f_h}{n_h} \, s^2_{\hat{z},h}, \\ \hat{z}_i &= \nabla f(\hat{T})^\top t_i(Y_i), \end{aligned} \quad (11)$$

$$V(\theta; n) = \sum_{h=1}^{H} N_h^2 \, \frac{1 - f_h}{n_h} \, S^2_{z,h} = \sum_h c_h S^2_{z,h}. \quad (12)$$

Proxy means and the difference estimator. Let $\pi_i \in [0, 1]$ be any proxy for $P(Y_i = 1)$ that is a function of $(\hat{y}_{mi}, \hat{p}_{mi}, G_i)_{i \in U}$ only (the design uses $\pi_i = M^{-1} \sum_m \hat{p}_{mi}$). Its *proxy mean* and the residual are

$$\begin{aligned} \psi_i &= \pi_i u_i + (1 - \pi_i) v_i, \\ \xi_i &= t_i(Y_i) - \psi_i = (Y_i - \pi_i)(u_i - v_i). \end{aligned} \quad (13)$$

Every unit of the pool carries a prediction, so ψ_i is known for all $i \in U$ and $F = \sum_{i \in U} \psi_i$

is a *known constant*: unlike prediction-powered inference with an unlabelled sample from a super-population [16, 57], there is no sampling error in the proxy total. For any label-free vector ψ_i the *difference estimator* is

$$\begin{aligned}\hat{T}_\psi &= \sum_{i\in U}\psi_i + \sum_{h=1}^{H} N_h\big(\bar{t}_{s_h} - \bar{\psi}_{s_h}\big)\\ &= \sum_{h=1}^{H}\big\{F_h + N_h\bar{\xi}_{s_h}\big\}, \qquad (14)\\ F_h &= \sum_{i\in U_h}\psi_i, \qquad \hat{\theta}_\psi = f(\hat{T}_\psi).\end{aligned}$$

With $\psi \equiv 0$ it is the plain estimator; with the proxy mean (13) it is the estimator of PATTERN-DIF, a stratified generalised difference estimator [17, 73]. The residual influence value, its sample version and the linearised variance are

$$\begin{aligned}e_i &= \nabla f(T)^\top \xi_i = z_i - \tilde{z}_i,\\ \tilde{z}_i &= \nabla f(T)^\top \psi_i,\\ \hat{V}_\psi &= \sum_h N_h^2 \frac{1-f_h}{n_h} s^2_{\hat{e},h},\\ \hat{e}_i &= \nabla f(\hat{T}_\psi)^\top \xi_i,\\ V_\psi(\theta;n) &= \sum_h c_h S^2_{e,h}. \qquad (15)\end{aligned}$$

The plain influence value splits into the part $\tilde{z}_i$ that the proxy predicts and the residual e_i that the labels must supply; $V_0 = V(\theta;n)$ is the plain case $\tilde{z} \equiv 0$.

5.2 Unbiasedness, consistency and asymptotic normality

Asymptotics refer to a sequence of pools with $N \to \infty$ and a fixed number H of strata. Let $r^2 = \sum_h W_h^2/n_h$, the order of $\mathrm{Var}(\hat{\tau}_c)$ for a generic total, and let ψ be 0, a proxy mean, or any label-free vector with $\|\psi_i\|_\infty \le 1$.

Assumption(A1) *Design.* Strata, sample sizes and the proxy are functions of $(\hat{y}_{mi}, \hat{p}_{mi}, G_i)_{i\in U}$ only (not refitted on the labelled sample); $2 \le n_h \le N_h$, $\min_h n_h \to \infty$, and $f_h \le 1-\varepsilon$ in every stratum that is not fully enumerated.

(A2) *Limits.* $\tau \to \tau_\infty$ and $P_g/N, Q_g/N \ge c > 0$ for $g \in \{0,1\}$.

(A3) *Non-degeneracy.* $V_\psi(\theta;n) \ge c\,r^2$ for some $c > 0$.

(A4) *Lindeberg-type condition.* In every stratum, either $n_h(1-f_h)S^2_{e,h}N^2 \to \infty$, or the stratum's share of $V_\psi(\theta;n)$ tends to zero.

(A1) excludes the use of labels in the design and the proxy; (A3) excludes degenerate estimands (for example two models with identical predictions) and, for the difference estimator, an essentially exact proxy; (A4) is a simple sufficient form of Hájek's [74] condition for the central limit theorem under sampling without replacement.

Proposition 1 (Validity) *Let $\hat{T}_\psi$, $\hat{\theta}_\psi$, $\hat{V}_\psi$ and $V_\psi(\theta;n)$ be as in (14)–(15); $\psi \equiv 0$ gives the plain estimator.*

(a) If $n_h \ge 1$ for all h, then $E(\hat{T}_\psi) = T$ and $\mathrm{Cov}(\hat{T}_\psi) = \sum_h N_h^2(1-f_h)n_h^{-1}\Sigma_h^\xi$, with Σ_h^ξ the stratum covariance matrix of ξ_i; if $n_h \ge 2$ the sample covariance matrix gives an exactly unbiased estimator. This holds for any label-free strata, allocation and ψ, in particular for any proxy.

(b) Under (A1)–(A2), $\hat{\theta}_\psi - \theta = O_p(r)$.

(c) Under (A1)–(A4), $(\hat{\theta}_\psi - \theta)/V_\psi(\theta;n)^{1/2} \to N(0,1)$.

(d) Under (A1)–(A3), $\hat{V}_\psi/V_\psi(\theta;n) \to 1$ in probability; under (A1)–(A4) the Wald interval $\hat{\theta}_\psi \pm z_{1-\alpha/2}\hat{V}_\psi^{1/2}$ has asymptotic coverage $1-\alpha$.

Proof sketch. (a) is the classical unbiasedness of the stratified expansion estimator applied to ξ_i, since $\hat{T}_\psi - T = \sum_h N_h(\bar{\xi}_{s_h} - \bar{\xi}_h)$ and the known F cancels. For (b)–(d), a second-order Taylor expansion of ϕ around τ (valid by (A2)) reduces $\hat{\theta}_\psi - \theta$ to a sum of H independent stratum means plus an $O_p(r^2)$ remainder, which (A3) makes negligible; each mean is asymptotically normal by Hájek's theorem, and consistency of the stratum sample covariances gives (d). Full proofs are in Supplementary Section S1. □

The estimator of a ratio-type estimand has bias $O(r^2)$, of smaller order than its standard deviation by (A3). The proof uses only that the

contributions are bounded and the proxy is label-free; it makes no use of the proxy being accurate or calibrated.

5.3 Structure of the influence values

For a pair (m, j) let $D_i = \hat{y}_{mi} - \hat{y}_{ji} \in \{-1, 0, 1\}$, $\delta_g = TPR_{mg} - TPR_{jg}$ and $\gamma_g = FPR_{mg} - FPR_{jg}$, so that $\theta^{TPR}_{mj} = \delta_1 - \delta_0$ and $\theta^{FPR}_{mj} = \gamma_1 - \gamma_0$.

Proposition 2 (Influence values) *For every instance i with $G_i = g$, with $s_1 = 1$, $s_0 = -1$,*

$$z_i^{TPR} = s_g \frac{Y_i(D_i - \delta_g)}{P_g},$$
$$z_i^{FPR} = s_g \frac{(1 - Y_i)(D_i - \gamma_g)}{Q_g}, \tag{16}$$
$$z_i^{ERR} = \frac{D_i(1 - 2Y_i)}{N}.$$

For a stratum h inside group g let $\pi_h = \bar{Y}_h$ (the stratum prevalence, not the proxy π_i), $\rho_h = N_h^{-1} \sum_{U_h} |D_i|$ (the disagreement rate of the pair), $r_h = N_h^{-1} \sum_{U_h} Y_i |D_i|$, $c_h = N_h^{-1} \sum_{U_h} Y_i D_i$ (a stratum moment, not the weight c_h of (12)) and $\kappa_h = N_h/(N_h - 1)$. Then

$$S^2_{z^{TPR},h} = \frac{\kappa_h}{P_g^2}\Big\{(r_h - c_h^2) + \delta_g^2 \pi_h(1 - \pi_h) - 2\delta_g\, c_h(1 - \pi_h)\Big\}, \tag{17}$$
$$S^2_{z^{ERR},h} = \frac{\kappa_h}{N^2}\Big\{\rho_h - e_h^2\Big\},$$
$$e_h = N_h^{-1} \textstyle\sum_{U_h} D_i(1 - 2Y_i), \tag{18}$$

and $S^2_{z^{FPR},h}$ is (17) with Y replaced by $1 - Y$, P_g by Q_g, δ_g by γ_g. In a stratum where m and j agree on every instance (always the case for $K_i \in \{0, M\}$),

$$S^2_{z^{TPR},h} = \kappa_h \Big(\frac{\delta_g}{P_g}\Big)^2 \pi_h(1 - \pi_h),$$
$$S^2_{z^{FPR},h} = \kappa_h \Big(\frac{\gamma_g}{Q_g}\Big)^2 \pi_h(1 - \pi_h), \tag{19}$$
$$S^2_{z^{ERR},h} = 0.$$

The proof is a direct calculation (Supplementary Section S1). Equation (19) is the structural fact behind the design. Where all models agree, the numerators of the comparison estimands receive nothing and only the denominator term remains, with label variability $\pi_h(1 - \pi_h)$ scaled by the square of the group-wise target component. Competitive models have $|\delta_g|$ of a few hundredths, so these strata, which contain most of the pool in seven of our eleven settings, carry a per-instance variance that is typically at least two orders of magnitude below that of the disagreement strata (for pairs among the three most accurate models on our real pools), where (17) is dominated by r_h. For the error difference the agreement strata carry no variance at all.

The residual influence values of the difference estimator are $e_i = (Y_i - \pi_i)d_i$ with $d_i = \nabla f(T)^\top (u_i - v_i)$ label-free (Proposition S2 in Supplementary Section S2); in particular $e_i^{ERR} = 0$ wherever the pair agrees. If the proxy is calibrated in the sense that the Y_i are independent Bernoulli(π_i), then $E_\pi S^2_{e,h} = \bar{\sigma}^2_h = N_h^{-1} \sum_{U_h} \pi_i(1 - \pi_i) d_i^2$ and $E_\pi S^2_{z,h} = S^2_{\tilde{z},h} + \bar{\sigma}^2_h$.

5.4 The difference estimator against the plain estimator

Let $\beta_h = \mathrm{Cov}_h(z, \tilde{z})/S^2_{\tilde{z},h}$ be the within-stratum regression slope of the plain influence value on its proxy mean, and R_h their within-stratum correlation, so $\beta_h = R_h S_{z,h}/S_{\tilde{z},h}$.

Proposition 3 (Variance of the difference estimator, design-based) *For any label-free proxy and any allocation, $S^2_{e,h} = S^2_{z,h} - 2\,\mathrm{Cov}_h(z, \tilde{z}) + S^2_{\tilde{z},h}$, hence*

$$V_0 - V_\psi = \sum_h c_h\, S^2_{\tilde{z},h}\,(2\beta_h - 1). \tag{20}$$

The difference estimator has the smaller linearised variance if and only if the right side is positive; a sufficient condition is $\beta_h > 1/2$ in every stratum. Further parts (optimal coefficient, calibrated form) are Proposition S3 in Supplementary Section S2.

The sign of the gain is therefore not automatic: the regression slope of z on $\tilde{z}$ must exceed one half, and a weakly aligned or over-dispersed proxy hurts. With a calibrated proxy the gain is guaranteed and computable without labels.

Proposition 4 (Design-effect ratio under a calibrated proxy) *Let the labels be independent Bernoulli(π_i) (expectation E_π), $\nu_h = S^2_{\tilde{z},h}/\bar{\sigma}^2_h$ the signal-to-noise ratio of the proxy influence in stratum h, and $a_h = c_h \bar{\sigma}^2_h$. Then*

(a) $E_\pi V_0 - E_\pi V_\psi = \sum_h c_h S^2_{\tilde{z},h} \geq 0$, with equality iff $\tilde{z}$ is constant within every stratum that is not fully enumerated.

(b) For the same strata and sample sizes,

$$\frac{E_\pi V_\psi}{E_\pi V_0} = \frac{1}{G},$$
$$G = \frac{\sum_h a_h\,(1+\nu_h)}{\sum_h a_h} = \frac{\sum_h a_h/(1-R_h^2)}{\sum_h a_h}, \tag{21}$$

where $R_h^2 = \nu_h/(1+\nu_h)$ is the expected within-stratum squared correlation of z_i with $\tilde{z}_i$.

The gain $G \geq 1$ is the average of the stratum gains $1/(1-R_h^2)$, weighted by the strata's shares of the residual variance, so a calibrated proxy never makes the difference estimator worse in expectation. The gain is a *within*-stratum R^2 and comes mostly from the known structure of u_i, v_i (the pattern of D_i), not from sharp proxies. Closed forms for $\theta^{TPR}, \theta^{FPR}, \theta^{ERR}$ (Proposition S4) give gains of the order of one to three in disagreement strata, for example 1.4, 3.3 and 1.2 for a constant proxy $\bar{\pi} = 0.3$, and they are multiplicative with the stratification gain. Proofs are in Supplementary Section S1.

Misspecified or tuned difference estimator. Unbiasedness and consistency of $\hat{V}_\psi$ hold for every proxy (Proposition 1). The loss from a wrong proxy is bounded: in all cases $V_\psi \leq (1 + \max_h S_{\tilde{z},h}/S_{z,h})^2 V_0$, and if the proxy influence is κ times the true one within strata, the difference estimator wins iff $0 < \kappa < 2$, so an anti-correlated or over-dispersed proxy loses to the plain estimator (Proposition S5). There is no guarantee against simple random sampling for an arbitrary proxy. Estimating the coefficient per stratum (Proposition S6) is asymptotically never worse than the plain estimator but adds $O(1/n_h)$ variance and a downward-biased variance estimator at small n_h; we therefore use the fixed coefficient $\lambda = 1$, which has no estimated parameters.

Remark 1 (Zero-variance strata and the variance floor) The Wald interval needs $\hat{V}_\psi > 0$ and fails when a stratum's sample shows no variation in $\hat{e}_i$ although $S^2_{e,h} > 0$ (for example a stratum with few positives that the sample misses). Structural zeros ($d_i \equiv 0$, e.g. the error difference where the pair agrees) are exact and harmless. For the other strata the design replaces $s^2_{\hat{e},h}$ by the label-free anticipated residual variance $\tilde{s}^2_h = N_h^{-1}\sum_{U_h} \pi_i(1-\pi_i)\hat{d}_i^2$ whenever $s^2_{\hat{e},h} = 0$ (equation S6). The floor never decreases $\hat{V}_\psi$, leaves the point estimate unchanged and does not affect Proposition 1(c)–(d) provided it is asymptotically negligible (Assumption (A5) of Section 6); it is a finite-sample safeguard, not an asymptotic correction (Supplementary Section S2).

5.5 Allocation

Let $w_t \geq 0$ be weights on a finite set of targets t with linearised variances $V_t(n) = \sum_h N_h^2(1/n_h - 1/N_h)S^2_{ht}$, where $S^2_{ht} = S^2_{z,ht}$ for the plain estimator and $S^2_{ht} = S^2_{e,ht}$ for the difference estimator, and let $A_h = (\sum_t w_t S^2_{ht})^{1/2}$. The weighted objective is

$$\Phi(n) = \sum_t w_t V_t(n) = \sum_h \frac{N_h^2 A_h^2}{n_h} - \sum_h N_h A_h^2. \tag{22}$$

Proposition 5 (Weighted Neyman allocation)*(a) Over real $n_h > 0$ with $\sum_h n_h = L$, Φ is minimised by $n_h^* = L\,N_h A_h/\sum_k N_k A_k$, with $\Phi(n^*) = (\sum_h N_h A_h)^2/L - \sum_h N_h A_h^2$.*

(b) With bounds $\ell_h \leq n_h \leq N_h$, the minimiser is $n_h = \min\{N_h, \max(\ell_h, N_h A_h/\lambda)\}$ with $\lambda > 0$ chosen so that $\sum_h n_h = L$.

(c) If an integer allocation $\tilde{n}$ satisfies $\tilde{n}_h \geq n_h - 1$ and $n_h \geq n_0 > 1$, then $\Phi_0(\tilde{n}) \leq \{n_0/(n_0-1)\}\Phi_0(n)$, with $\Phi_0(n) = \sum_h N_h^2 A_h^2/n_h$.

(d) Ignoring finite-population corrections, with $w_t = 1/\sigma_t^2$ (σ_t^2 the population variance of the influence values of target t), $\Phi_0(n)\,L/N^2 = \sum_t 1/\mathrm{Eff}_t(n)$ with $\mathrm{Eff}_t = V_t^{SRS}/V_t$; at the optimum the harmonic mean efficiency over $|\mathcal{T}|$ targets equals $|\mathcal{T}|/(\sum_h W_h A_h)^2 \geq 1$.

Proof sketch. (a) is Neyman's allocation [65] for a weighted sum of variances: by Cauchy–Schwarz, $(\sum_h N_h A_h)^2 \leq \Phi_0(n)\,L$, with equality iff $n_h \propto N_h A_h$. (b) follows from the Karush–Kuhn–Tucker conditions of a convex problem (strata that would exceed N_h are enumerated and the rest re-allocated). Full proofs are in Supplementary Section S1. □

Because the S^2_{ht} are unknown, the design replaces them by their expectations under the

working model $Y_i \sim \text{Bernoulli}(\pi_i)$ independently [the anticipated variance of 75], which by the moments above gives

$$\begin{aligned} \tilde{A}_h^{0\,2} &= \sum_t w_t\big(\bar{\sigma}_{ht}^2 + S_{\tilde{z},ht}^2\big) \text{ (plain rule)}, \\ \tilde{A}_h^{\psi\,2} &= \sum_t w_t \bar{\sigma}_{ht}^2 \text{ (difference rule)}. \end{aligned} \tag{23}$$

The design uses the plain rule (Section 4), which is exactly Neyman for the plain estimator and does not depend on trusting the proxy. For the difference estimator the loss from using the plain rule is at most $\{1+(1+\nu_{\max})^{1/2}\}^2/\{4(1+\nu_{\max})^{1/2}\}$ (for example 9/8 for $\nu_{\max}=3$), by Proposition S7 in Supplementary Section S2.

5.6 Design effect and the disagreement rate

With σ_{ht}^2 and σ_t^2 the variances with divisors N_h and N, a single target allocated by its own Neyman rule with $L/N \to 0$ has efficiency relative to SRS

$$\text{Eff}_t = \frac{\sigma_t^2}{(\sum_h W_h \sigma_{ht})^2} \;\geq\; 1, \tag{24}$$

with equality only if all within-stratum standard deviations are equal and the stratum means coincide. Combined with Proposition 2, this gives explicit formulas (Corollary S8 in Supplementary Section S2). They reduce to the heuristic *efficiency* $\approx 1/\rho$, where ρ is the pair's disagreement rate: it is exact for an error difference with a two-sided disagreement set, and holds up to a group factor and a bounded target-size factor for the rate differences. A pair that disagrees on 5% of the pool gains about 20 over SRS, one that disagrees on 40% about 2.5. For $M > 2$ pattern strata give $1/\rho_{\text{any}} \leq \text{Eff}^{ERR} \leq 1/\rho$ (equation S13). The difference estimator multiplies these stratification gains by the factor G of (21), which is of order one to three and does not grow as $\rho \to 0$.

5.7 Misspecified proxy

The proxy enters the design only through the anticipated stratum variances $\tilde{A}_h$. Validity (Proposition 1) holds whatever the proxy, provided $\tilde{n}_h \geq 2$. The relative loss in the weighted variance from using $\tilde{A}_h$ instead of the true A_h is $(\sum_h \omega_h x_h)(\sum_h \omega_h/x_h) \geq 1$ with $\omega_h = N_h A_h / \sum_k N_k A_k$ and $x_h = \tilde{A}_h/A_h$; it vanishes iff $\tilde{A}_h \propto A_h$ (a proxy wrong by a common factor costs nothing) and is bounded by the Kantorovich ratio $(x_{\max}+x_{\min})^2/(4x_{\max}x_{\min})$, so if every $\tilde{A}_h$ is within a factor 2 of A_h up to a common scale the design keeps at least 64% of the efficiency of the optimal allocation (Proposition S9). What cannot be guaranteed is improvement over SRS for every population: a proxy that is badly wrong in a high-variance stratum can do worse than proportional allocation. A defensive mixture $n_h = \eta\, n_h^{\text{prop}} + (1-\eta)\tilde{n}_h$ guarantees a weighted variance at most $1/\eta$ times that of SRS when the sampling fraction L/N is small (up to a factor $(1-1/\min_h N_h)^{-1}$); the exact factor with finite-population corrections is $(1-\eta f)/\{\eta(1-f)\}$ for $f = L/N$ (Supplementary Proposition S9). Which strata carry the numerators is determined by the predictions, not by the proxy (Proposition 2).

5.8 Numerical check of the theory

The design effects of this section are first-order (linearised) quantities, so we checked them against Monte Carlo experiments on the eleven real pools and on synthetic pools in which the disagreement rate is controlled (Supplementary Section S5). We report three findings. First, the linearised design effect predicts the efficiency of the implemented design to within Monte Carlo error: over all target–budget pairs on the real pools the ratio of observed to predicted efficiency has median 1.00 (10%–90% range 0.83–1.22). Second, the closed-form expression of Corollary S8 is accurate when the strata separate the agreements of a pair from its disagreements (two models), and captures the $1/\rho$ scaling and the effect of the size of the contrast; with nine models it is too optimistic and overstates the gain on real pools by a median factor of about two, because the count K_i locates any disagreement but not that of a particular pair, and the bounds of Supplementary Section S2 describe the gap. We therefore treat the closed form as an illustration of the mechanism, not as a forecast. Third, for the difference estimator the linearised residual variance predicts its gain over the plain estimator within Monte Carlo error (median ratio of observed to predicted gain 0.98 over 34 pool–budget cells, Supplementary Section S5.4); the gain is moderate (median 1.47) and is largest in the law-school setting, where

the label-free forecast underestimates it. An uninformative or inverted proxy leaves the allocation almost unchanged, and a proxy of the wrong sign makes the difference estimator worse than the plain one for the error contrast by a factor of about 1.3, as the theory predicts.

6 Hypothesis tests for fairness contrasts

Section 5 gives estimators and intervals. A user also has to decide whether model m has a smaller TPR gap than model j, whether a candidate is no worse than the incumbent by more than a margin δ, and which of the $Q = 3\binom{M}{2}$ contrasts (Q is used to avoid the stratum variable K_i) differ from zero when all are asked at once. This section gives the tests, their size and power, the simultaneous procedures and the design that maximises power, with the assumptions under which each holds. Proofs and the remarks omitted here are in Supplementary Section S3.

A hypothesis is a statement about a contrast of the pool, $\theta_k = f_k(T)$. The only randomness is the labelling lottery, so a p-value is a probability over that lottery for the fixed models and pool; it says nothing about retraining (Section 7.6 treats the population contrast). Write $\hat{\theta}_k = f_k(\hat{T}_\psi)$, $V_k = V_\psi(\theta_k; n)$, $\hat{V}_k$ for the quantities of (14)–(15), and

$$Z_k(\theta_0) = \frac{\hat{\theta}_k - \theta_0}{\hat{V}_k^{1/2}}, \qquad z_q = \Phi^{-1}(q). \tag{25}$$

All limits are along a sequence of pools as in Section 5.2; contrasts with $V_k = 0$ are excluded by (A3) and removed from the family.

(A5) *Floor.* If the variance floor of Remark 1 is used, $\hat{V}_k^\flat / \hat{V}_k \to 1$ in probability, with $\hat{V}_k^\flat \geq \hat{V}_k$ the floored estimator.

(A4$_Q$) *Joint Lindeberg condition.* (A4) holds for every linear combination of the Q residual influence vectors whose variance is of exact order r^2.

(P1) *Population.* The pool is a simple random sample of N units (X_i, G_i, Y_i) from a fixed infinite population; models, strata rule and proxy rule are label-free functions of (X, G).

(P2) *Regularity.* Population versions of (A2) hold ($P(G = g, Y = y) \geq c$), the population influence value has variance $\sigma_\zeta^2 > 0$, and with probability tending to one the pool lies in a class $\mathcal{P}_N$ of regular pools: $n_h \geq m_N \to \infty$ in every stratum, $f_h \leq 1 - \varepsilon$ outside fully enumerated strata, $V_\psi(\theta; n) \geq c\, r^2$, and each stratum either has $n_h(1 - f_h) S_{e,h}^2 N^2 \geq 1/\eta_N$ or a share at most $\eta_N \to 0$ of V_ψ (complete statement in Supplementary Section S3.10). Every sequence from $\mathcal{P}_N$ then satisfies (A1)–(A4).

(P3) *Concentration.* There is $v_{\mathrm{D}} > 0$ with $V_\psi(\theta; n)/v_{\mathrm{D}} \to 1$ in probability over the pool.

(A5) and (P3) are high-level conditions on the design; they are discussed at the end of the section.

6.1 Wald tests and their size

For a null value θ_0 (asymptotic Wald tests are standard, Lehmann and Romano 76, Ch. 11; what is new is the design-based setting) the two-sided test rejects $H_0 : \theta_k = \theta_0$ when $|Z_k(\theta_0)| > z_{1-\alpha/2}$, the upper one-sided test rejects $H_0 : \theta_k \leq \theta_0$ when $Z_k(\theta_0) > z_{1-\alpha}$, and the lower one is symmetric. For a margin $\delta > 0$ the non-inferiority test of $H_0 : \theta_k \geq \delta$ rejects when $Z_k(\delta) < -z_{1-\alpha}$. The equivalence test of $H_0 : |\theta_k| \geq \delta$ is the two one-sided tests (TOST) procedure [77, 78], an intersection-union test [79, 80]: it rejects when $Z_k(-\delta) > z_{1-\alpha}$ and $Z_k(\delta) < -z_{1-\alpha}$, i.e. when the $(1 - 2\alpha)$ Wald interval lies inside $(-\delta, \delta)$ (so $\alpha < 1/2$).

Theorem 6 (Size) *Let $0 < \alpha < 1/2$, and consider a sequence of pools with (A1)–(A4) for θ_k (and (A5) if the floor is used), with $\theta_0 = \theta_{0,N}$, $\delta = \delta_N > 0$.*

(a) If $\theta_k = \theta_0$, then $Z_k(\theta_0) \to N(0, 1)$ in law, the two-sided test has rejection probability tending to α, and $2\{1 - \Phi(|Z_k|)\}$ is asymptotically uniform.

(b) If $\theta_k \leq \theta_0$, the upper one-sided test has $\limsup P(\text{reject}) \leq \alpha$, with limit α when $(\theta_k - \theta_0)/V_k^{1/2} \to 0$.

(c) If $\theta_k \geq \delta$, the non-inferiority test has $\limsup P(\text{reject}) \leq \alpha$. If $|\theta_k| \geq \delta$ the TOST has $\limsup P(\text{reject}) \leq \alpha$, with limit α when $|\theta_k| = \delta$ and $\delta/V_k^{1/2} \to \infty$.

(d) *(Population target.) Under (P1)–(P3), let* $v_{\mathrm{P}} = \sigma_\zeta^2/N$, $s^2 = v_{\mathrm{D}} + v_{\mathrm{P}}$, θ_{pop} *the population contrast and* $Z^{\mathrm{pop}} = (\hat\theta_k - \theta_0)/(\hat V_{\mathrm{design}} + \hat V_{\mathrm{pool}})^{1/2}$ *with* $\hat V_{\mathrm{pool}}$ *of Section 7.6. If* $\theta_{\mathrm{pop}} = \theta_0$, *(a)–(c) hold for* Z^{pop}, *and the power statements of Theorem 7 hold with* $\mu = (\theta_{\mathrm{pop}} - \theta_0)/s$. *The pool-conditional* Z_k *applied to the same null has limiting rejection probability* $2\{1-\Phi(z_{1-\alpha/2}/\sqrt{1+\varpi})\}$ *when* $\varpi = v_{\mathrm{P}}/v_{\mathrm{D}} \to \varpi_\infty$, *which exceeds* α *unless* $\varpi_\infty = 0$.

Proof sketch. Write $Z_k(\theta_0) = (X+\mu)/\varrho$ with $X \to N(0,1)$ by Proposition 1(c), $\varrho = (\hat V_k/V_k)^{1/2} \to 1$ by Proposition 1(d) and $\mu = (\theta_k - \theta_0)/V_k^{1/2}$. A Pólya–Slutsky argument gives $P(X+\mu > z\varrho) = \Phi(\mu - z) + o(1)$ uniformly in μ; (a)–(c) follow at $\mu = 0$ or by monotonicity, and TOST is the minimum of its two components. For (d), a characteristic-function argument that uses (P3) shows that the design and pool errors add to a normal with variance $v_{\mathrm{D}} + v_{\mathrm{P}}$. □

Ignoring the pool term in (d) inflates the size to 0.07 at $\varpi = 0.2$ and to 0.17 at $\varpi = 1$ for $\alpha = 0.05$; the statement is asymptotic.

6.2 Power, local alternatives and Pitman efficiency

For a fixed pool θ_k is a number, so the power of a test of $H_0 : \theta_k = \theta_0$ is a function of the hypothesised value θ_0, and "local" means $|\theta_k - \theta_0| = O(V_k^{1/2})$.

Theorem 7 (Power curve) *Under the assumptions of Theorem 6, with* $\mu = (\theta_k - \theta_0)/V_k^{1/2}$, $z = z_{1-\alpha/2}$ *and* $\Psi_\alpha(\mu) = \Phi(\mu - z) + \Phi(-\mu - z)$,

$$\sup_{\theta_0 \in \mathbb{R}} \left| P\big(|Z_k(\theta_0)| > z\big) - \Psi_\alpha(\mu) \right| \to 0. \qquad (26)$$

The one-sided tests have power $\Phi(\pm\mu - z_{1-\alpha})$ *and the TOST* $[\Phi(b - z_{1-\alpha}) - \Phi(z_{1-\alpha} - a)]_+$ *with* $a = (\theta_k + \delta)/V_k^{1/2}$, $b = (\delta - \theta_k)/V_k^{1/2}$, *uniformly. In particular the tests are consistent.*

Proof sketch. The uniform Slutsky argument of Theorem 6 applied to $\{X+\mu > z\varrho\}$ and $\{X+\mu < -z\varrho\}$. □

Power depends on the design only through V_k, and Ψ_α increases in $|\mu|$, so comparing designs is comparing variances. A design allocating $n_h = \lambda_h L$ labels with fixed shares λ_h has $V^{(d)}(L) = v_d/L - w_d$ with $v_d = \sum_h N_h^2 S_h^2/\lambda_h$ and $w_d = \sum_h N_h S_h^2$, where S_h^2 are the stratum variances of the influence values (residual values for the difference estimator), valid while $\lambda_h L \le N_h$. SRS is the one-stratum plain design, and $\mathrm{Eff} = V^{\mathrm{SRS}}(L)/V^{\mathrm{PD}}(L)$.

Corollary 8 (Labels needed and efficiency) *Let* $\mu_{\alpha,\beta}$ *be the positive root of* $\Psi_\alpha(\mu) = 1-\beta$; $\mu_{\alpha,\beta} \le z_{1-\alpha/2} + z_{1-\beta}$, *with equality* $z_{1-\alpha} + z_{1-\beta}$ *for one-sided tests. Under the assumptions of Theorem 6, a design with* $\lambda_h L_d \le N_h$ *reaches asymptotic power* $1-\beta$ *against* $|\theta_k - \theta_0| = \Upsilon$ *if and only if* $L \ge L_d$, *where*

$$L_d(\Upsilon) = \frac{\mu_{\alpha,\beta}^2\, v_d}{\Upsilon^2 + \mu_{\alpha,\beta}^2\, w_d}. \qquad (27)$$

Hence for two designs

$$\frac{L_2}{L_1} = \frac{v_2}{v_1} \cdot \frac{\Upsilon^2 + \mu_{\alpha,\beta}^2 w_1}{\Upsilon^2 + \mu_{\alpha,\beta}^2 w_2} \longrightarrow \frac{v_2}{v_1} = \lim \frac{V^{(2)}(L)}{V^{(1)}(L)}, \qquad (28)$$

the limit holding when the sampling fractions $\lambda_h L_d/N_h$ *tend to zero. For PATTERN-DIF against SRS this limit is* Eff. *For the population target* v_{P} *is added to the variance, so power is capped at every* L: *no budget reaches* $1-\beta$ *unless* $\Upsilon > \mu_{\alpha,\beta}\, v_{\mathrm{P}}^{1/2}$.

Proof sketch. Equal power means equal $\mu = \Upsilon/V^{1/2}$, hence $V^{(d)}(L_d) = \Upsilon^2/\mu^2$. □

Two qualifications apply: the claim is asymptotic (small labelled fraction), and for the population contrast it concerns labels for a given power, not the mean squared error at a given budget, whose ratio is $(\mathrm{Eff} + \varpi)/(1+\varpi)$ (Section 7.6). In practice v_d, w_d are replaced by anticipated values (23), so (27) is a planning tool as accurate as those.

6.3 Joint inference over many contrasts

Let $\Sigma_{kl} = \sum_h c_h \operatorname{Cov}_h(e^{(k)}, e^{(l)})$ be the covariance of the linearised estimators and R the correlation matrix, estimated by $\hat\Sigma$ (floor added to the diagonal) and $\hat R$. The paper's family is singular: $\theta_{ml} = \theta_{mj} + \theta_{jl}$ for each type, so the $Q = 108$ contrasts of $M = 9$ models are linear images of $3(M-1) = 24$ gaps and R has rank at most 24; the theory allows singular R.

Theorem 9 (Joint asymptotics) *Let (A1)–(A3) hold for each of the Q contrasts (fixed Q), together with ($A4_Q$) and (A5). Then $\hat{V}_k/V_k \to 1$, $\hat{R} - R \to 0$ in probability, and $\{(\hat{\theta}_k - \theta_k)/V_k^{1/2}\}_{k \le Q}$ is asymptotically $N(0, R)$: along any subsequence with $R \to R_\infty$ it converges in law to $N(0, R_\infty)$.*

Proof sketch. Proposition 1(b) linearises every contrast with an $O_p(r^2)$ remainder; for a fixed linear combination the argument of Proposition 1(c) applies under ($A4_Q$), and Cramér–Wold gives joint convergence. Consistency of $\hat{\Sigma}$ is that of Proposition 1(d), entry by entry (Supplementary Section S3.5). □

Let $c_\alpha(R)$ be the $(1-\alpha)$ quantile of $\max_{k \le Q} |N_k|$, $N \sim N(0, R)$ [81, 82]. The *max-$|t|$ procedure* rejects $H_k : \theta_k = \theta_{k,0}$ when $|Z_k(\theta_{k,0})| > c_\alpha(\hat{R})$; its step-down version [83] recomputes the critical value on the hypotheses not yet rejected. Bonferroni rejects when $|Z_k| > z_{1-\alpha/(2Q)}$, and Holm's procedure [84] is its step-down version.

Theorem 10 (Familywise error and simultaneous bands) *Under the assumptions of Theorem 9, for every set I_0 of true hypotheses,*

(a) max-$|t|$, its step-down version, Bonferroni and Holm each reject at least one true H_k with probability at most $\alpha + o(1)$ (strong FWER control), and single-step max-$|t|$ has limit exactly α when $I_0 = \{1, \dots, Q\}$;

(b) the band $\hat{\theta}_k \pm c_\alpha(\hat{R}) \hat{V}_k^{1/2}$, $k \le Q$, covers all θ_k simultaneously with probability tending to $1-\alpha$, hence also any contrast selected after looking at the data;

(c) $c_\alpha(R) \le z_{1-\alpha/(2Q)}$ and $c_\alpha(R_S) \le c_\alpha(R)$ for every subset S; so single-step (step-down) max-$|t|$ rejects every hypothesis that Bonferroni (Holm) rejects, for every data set.

Proof sketch. $R \mapsto c_\alpha(R)$ is continuous on correlation matrices. For $k \in I_0$ the statistics do not involve the false nulls, and $\max_{I_0} |Z_k| \to \max_{I_0} |N_k|$ with $c_\alpha(\hat{R}) \to c_\alpha(R_\infty) \ge c_\alpha(R_{\infty, I_0})$ (subset pivotality of the Gaussian limit), so a false rejection has asymptotic probability at most α; in the step-down version the active set contains I_0 until a true null is rejected. Part (c) is the union bound and monotonicity (Supplementary Section S3.7). □

The gain over Bonferroni comes from correlation between contrasts that share a model. The band answers the multiplicity and selection concern of Step 5 of Section 10: a statement about the best pair found after inspecting all of them is covered.

6.4 Designing the allocation for power

Let S_{hk}^2 be the stratum variance of the residual influence values of contrast k (anticipated in practice), $V_k(n) = \sum_h N_h^2 (1/n_h - 1/N_h) S_{hk}^2$, $\delta_k > 0$ the effect to be detected, $\mu_k(n) = \delta_k / V_k(n)^{1/2}$, and $\mathcal{N} = \{n : \sum_h n_h = L,\ \ell_h \le n_h \le N_h\}$. For $\omega \in \mathbb{R}^Q_{\ge 0}$ and $A_h(\omega)^2 = \sum_k \omega_k S_{hk}^2$ the weighted Neyman allocation of Proposition 5(b) is

$$n_h(\omega) = \min\{N_h, \max(\ell_h, N_h A_h(\omega)/\lambda)\}, \quad (29)$$

with $\lambda > 0$ such that $\sum_h n_h(\omega) = L$.

Theorem 11 (Optimal design for power) *(a) For one contrast the Neyman allocation minimises V_k and so maximises the power (26) for every effect size at once. (b) The maximin problem $\min_{n \in \mathcal{N}} \max_k V_k(n)/\delta_k^2$ is a convex program with a saddle point; if w^* maximises the concave dual $\Gamma(w) = \min_{n \in \mathcal{N}} \sum_k w_k V_k(n)/\delta_k^2$ over the simplex and $A_h(\omega^*) > 0$ for all h, the optimal allocation is unique and equals $n(\omega^*)$ with $\omega_k^* = w_k^*/\delta_k^2$, i.e. $n_h^* \propto N_h \{\sum_k w_k^* S_{hk}^2/\delta_k^2\}^{1/2}$ before bounds, and only contrasts with maximal relative variance V_k/δ_k^2 get positive weight. (c) The default weights $\omega_k^{\mathrm{def}} = 1/\sigma_k^2$ minimise the sum of inverse efficiencies, not $\max_k V_k/\delta_k^2$, and are in general not power-optimal; they are maximin-optimal if $V_k(n^{\mathrm{def}})/\delta_k^2$ is the same for all k, and $\max_k V_k(n^{\mathrm{def}})/\delta_k^2 / \Gamma(w)$ bounds the loss for any w. (Part (d), on integer rounding, and the full statement are in Supplementary Section S3.9.)*

Proof sketch. (a) Ψ_α is increasing in $|\mu|$. (b) V_k is convex in n and the objective linear in w, so Sion's theorem [85] gives a saddle point whose inner minimiser is the weighted Neyman allocation. (c) is a direct computation. □

Under max-$|t|$ the power of each contrast is at least $\Phi(\min_k \mu_k - z_{1-\alpha/(2Q)})$, a design-free bound that the maximin design maximises; it does not maximise the power itself, since $c_\alpha(R)$ depends on the allocation. The allocation of Section 4

is computed from plain variances, whereas S^2_{hk} above are those of the estimator being tested, so for PATTERN-DIF the results describe the residual-variance rule.

6.5 The equalised-odds gap

The equalised-odds gap $\mathrm{EO}_m = \max(|\Delta^{TPR}_m|, |\Delta^{FPR}_m|)$ is not differentiable at ties, so the delta method behind Proposition 1 does not apply there. Proposition S15 (Supplementary Section S3.11) shows that the plug-in is asymptotically normal away from ties, has a non-normal limit and an upward bias of order r near ties, and that the intersection-union test of $\mathrm{EO}_m < \delta$ has size at most α regardless of ties. The paper therefore tests the signed gaps, uses that test for margin questions, and reports EO_m rankings descriptively (Section 9).

6.6 What is and is not proved

Everything above is asymptotic with H and Q fixed, for pools satisfying (A1)–(A4). (A5) and (P3) are high-level statements that we do not derive from primitive assumptions: the first says the variance floor does not matter asymptotically, the second that the random pool does not move the design variance on the relative scale (it would follow from a law of large numbers for stratum moments and a continuous allocation rule, which we have not proved). ($\mathrm{A4}_Q$) is Hájek's condition for combinations of contrasts and cannot be replaced by the marginal conditions. The population-target size results rest on (P2) and (P3) and on the pool being a simple random sample with fixed models, so they do not cover a convenience pool or models refitted on the same data. Finite-sample size and power are not addressed by the theory and are checked by simulation; at small budgets the nominal levels are not reached for ratio contrasts with few sampled positives.

6.7 Size, power and familywise error: simulations

We check whether the Wald tests have nominal size, whether the simultaneous procedures control the familywise error rate (FWER) over the $K = 108$ contrasts, whether the label budgets from the power formula are the observed ones, and whether allocating labels for power helps. Protocol and full tables are in Supplementary Section S4.

Set-up.

The synthetic population is the generative model of the non-smooth simulation with nine fixed scoring rules (36 pairs, 108 contrasts). Four rules use a covariate that does not enter the label model and each has a twin using another exchangeable covariate, so the three contrasts of each of these four pairs are exactly zero in the population (12 true nulls) although the twins' predictions differ on much of every pool. We compare SRS, PATTERN and PATTERN-DIF with the variance estimators of Section 4, on pools of $N \in \{2000, 8000\}$ drawn anew in every replication, $L \in \{250, 500, 1000, 2000\}$ with $L \leq N/4$ (six cells) and $R = 1000$. The pool-conditional test targets the pool contrast, the two-phase test the population contrast (Section 7.6); on the eleven cached real pools the truth is the pool contrast. Simultaneous procedures use $\alpha = 0.05$ (max-t from 20,000 normal draws). The Monte Carlo standard error of a rejection rate near 0.05 is 0.007, and the largest of 108 such rates from a calibrated test has mean 0.068, so "max" columns are compared with this value and not with 0.05.

Size.

Table 1 and Figure 1(a) give the results for PATTERN-DIF. The two-sided Wald test of the true value rejects in 0.048 to 0.053 of replications on average at $\alpha = 0.05$ in all six cells, for both the pool-conditional test of the pool contrast and the two-phase test of the population contrast. The worst contrast is rejected in 0.063 to 0.074; the largest value is just above 0.068 and we do not read it as a failure. SRS and PATTERN behave alike. One-sided tests at the boundary reject in 0.048 to 0.055 and the TOST at the boundary is conservative (Supplementary Table S3). The pool term matters for the target, not for the test of the target chosen: the pool-conditional test of the *population* value rejects in 0.060 to 0.175 at nominal 0.05, increasing with ϖ, while the two-phase test stays at 0.05 (Supplementary Table S2).

Familywise error.

With only 12 true nulls among 108 hypotheses the FWER of every adjusted procedure is

Table 1 PATTERN-DIF, synthetic population, $R = 1000$ pools per row. Size: rejection rate of the two-sided Wald test of the true value at nominal 0.05 (pool-conditional test of the pool contrast; two-phase test of the population contrast), mean and maximum over the 108 contrasts (the maximum of a calibrated test has mean 0.068). ϖ: median estimated $V_{\text{pool}}/V_{\text{design}}$. FWER: probability of rejecting at least one of the 12 true nulls of $H_k : \theta_k = 0$, $k = 1, \ldots, 108$, at familywise level 0.05 (population target).

			Size, pool value		Size, population value		FWER, population nulls			
N	L	ϖ	mean	max	mean	max	Unadj.	Bonf.	Holm	Max-t
2,000	250	0.39	0.051	0.066	0.051	0.067	0.398	0.006	0.008	0.009
	500	1.11	0.053	0.069	0.051	0.069	0.406	0.007	0.008	0.008
8,000	250	0.08	0.052	0.069	0.051	0.066	0.353	0.005	0.006	0.009
	500	0.20	0.048	0.064	0.049	0.067	0.375	0.002	0.003	0.004
	1000	0.46	0.050	0.069	0.048	0.063	0.386	0.007	0.011	0.009
	2000	1.19	0.052	0.074	0.053	0.071	0.450	0.006	0.008	0.008

far below 0.05 (0.002 to 0.011 for Bonferroni, Holm and max-t, against 0.35 to 0.45 unadjusted; Table 1), which is conservative but uninformative about tightness. We therefore also test the family in which all 108 hypotheses are true, where the FWER is one minus the simultaneous coverage (Figure 1(b), Supplementary Table S4). For PATTERN-DIF the unadjusted tests reject at least one true hypothesis in 0.76 to 0.90 of replications; Bonferroni and Holm have FWER 0.03 to 0.05, and max-t 0.05 to 0.07 (mean 0.058 for both targets). Max-t is thus slightly anticonservative, by about one percentage point, consistent with a critical value only 3% below Bonferroni's (the contrasts are only moderately correlated and $\hat{R}$ adds noise) and the anticonservatism is larger for SRS at small budgets (up to 0.09). Holm and max-t are more powerful than Bonferroni by 0.01 to 0.03, and neither dominates the other.

Real pools.

On the eleven real pools (35 setting–budget cells, $R = 400$) the unadjusted PATTERN-DIF intervals cover all pool contrasts at once in only 0.20 to 0.33 of replications, and over all cells the mean coverage is 0.954 for Bonferroni and Holm and 0.923 for max-t (Monte Carlo standard error 0.011). On real pools max-t is thus *below* Bonferroni and below its level, by 0.03 on average, and the shortfall is concentrated: it is under 0.90 in four of the 35 cells (Law school at $L = 250$ and 500 with 0.65 and 0.89, MAGIC at $L = 250$ with 0.885, Spambase at $L = 250$ with 0.897) where the individual intervals also undercover at small budgets, so the failure is that of the intervals and not of the max-t construction; the plug-in estimators are much worse (Supplementary Tables S5 and S6).

Label budgets.

For six contrasts per setting we interpolated simulated power curves to the budget for 80% power and compared the ratio $L_{\text{SRS}}/L_{\text{DIF}}$ with the prediction from the power function with exact finite-pool variances. Figure 1(c) and Supplementary Table S7 show close agreement: over 49 contrasts in nine settings the median of observed over predicted is 1.00 (quartiles 0.97 and 1.02), all ratios lie within ±20% and the rank correlation is 0.99 (two contrasts censored). The plain asymptotic ratio of variances overstates the observed budget ratio by about 12% (observed about 0.88 times, range 0.78 to 1.08), because the finite-pool factor $1/L - 1/N$ makes the SRS variance fall faster than $1/L$ at the budgets reached; the formula that includes it is accurate.

Power-optimal allocation.

For the synthetic pool and three real pools (Adult (sex), Bank marketing, Electricity) with $L \in \{500, 1000\}$ we solved the maximin problem of Theorem 11 with two target sets, E1 (contrasts with $|\theta_k| \geq 0.02$ at effect $|\theta_k|$) and E2 (all contrasts at effect 0.05), *using the true contrasts as targets*, an idealisation. Compared with the default allocation, the maximin allocation reduces $\max_k V_k/\delta_k^2$ by a factor of 1.5 to 2.0 in the real pools and 1.05 on the synthetic pool. Figure 1(d) shows that worst-contrast power rises but from low levels (E2: 0.09 to 0.13 in Adult (sex), 0.17 to 0.23 in Bank marketing, 0.37 to 0.65 in Electricity), whereas *mean* power falls: in Adult (sex) at $L = 1000$ under E2 from 0.87 to 0.79, with contrasts below 80% rising from 21 to 35 (Supplementary Table S9). Power-optimal design helps

the worst-off contrasts at the cost of the others; it is not a free improvement.

Limitations.

The nulls are exact only in the population and there are only 12 of them, so the adjusted FWER with true nulls is conservative and the all-true family is the demanding check. There is one synthetic design and eleven real pools, and max-t is mildly anticonservative at small budgets. Further limitations are in Supplementary Section S4.

7 Empirical study on eleven data settings

7.1 Data, models and protocol

We use eleven settings built from ten public binary classification data sets available on OpenML (Table 2; details in Supplementary Section S9). In five settings the group is a protected attribute: sex or race in Adult, age below 30 in Bank Marketing, sex in the credit-card default data and race in the law-school admission data. In the other six the group is an arbitrary split of the first numeric feature at its median, removed from the features; these settings have no fairness meaning but cover a wider range of prediction structures. A twelfth candidate (Nomao) was excluded because its arbitrary group contains no positive instances.

Each data set is split once into a training part (60%) and a pool (40%); the Electricity data (45,312 rows) were first subsampled to 30,000, while Adult and Bank Marketing are used in full. Nine classifiers are trained on the training part with scikit-learn: logistic regression with default and with strong regularisation, decision trees of depth 3 and 12, a random forest, histogram gradient boosting, 25-nearest neighbours, Gaussian naive Bayes and a multilayer perceptron. Their predictions and predicted probabilities on the pool, together with the group attribute, are all the design uses. The pool labels play two roles: they answer label queries, and they define the true values of the estimands, so that the error of every estimate is known exactly.

For each setting and budget $L \in \{250, 500, 1000, 2000\}$ with $L \leq N/4$ (35 setting–budget cells), we draw 200 independent labelled samples under each design and compute the $3\binom{9}{2} = 108$ comparison estimands of (4). The efficiency of a design is the ratio of the mean squared error under SRS to that under the design, computed for each estimand and summarised by the median over estimands; an efficiency of 4 means that SRS needs about four times as many labels for the same accuracy. Estimands whose true value is identically degenerate in the pool (for example when no model pair disagrees in a group) are excluded; this affects 51 of 108 estimands in the law-school setting only. Efficiencies are computed over the samples in which both designs give finite estimates; SRS produced an undefined estimate (no sampled positive in a group) in at most 0.5% of its samples and the stratified designs never did. The designs are those of Section 4: SRS, SRS-DIF, GROUP, GROUP-DIF, PATTERN (plain estimator) and the proposed PATTERN-DIF, together with a variant of PATTERN-DIF whose allocation uses the residual variances. SRS, GROUP and PATTERN use the same random seeds as the earlier plug-in study (`results/label_efficiency.json`), so their samples coincide with it and their efficiencies agree except in law school, where that study used a different set of compared estimands. The code and all results are in the repository (`experiments/label_efficiency_dif.py`, `results/label_efficiency_dif.json`).

7.2 Efficiency

Table 4 gives the efficiency of every design in every cell, Table 3 summarises it by budget and Figure 2 shows it as a function of the budget and of the disagreement rate. Across settings, the median efficiency of PATTERN-DIF relative to SRS is 5.8 at $L = 250$ (eleven settings), 6.3 at $L = 500$ (ten), 6.9 at $L = 1000$ (eight) and 15.5 at $L = 2000$ (six); without the degenerate law-school setting the medians are 5.5, 5.8, 6.0 and 9.4. SRS thus needs between five and sixteen times as many labels to match PATTERN-DIF, and the lowest efficiency in any cell outside law school is 3.5. The medians over settings rise with the budget partly because the set of settings changes (only the larger pools allow the larger budgets): on the seven non-degenerate settings that are present at $L = 250$, 500 and 1000 the medians of PATTERN-DIF are 5.3, 5.3 and 6.0 (3.5, 3.8 and 4.4 for PATTERN). Within each of these seven settings the efficiency

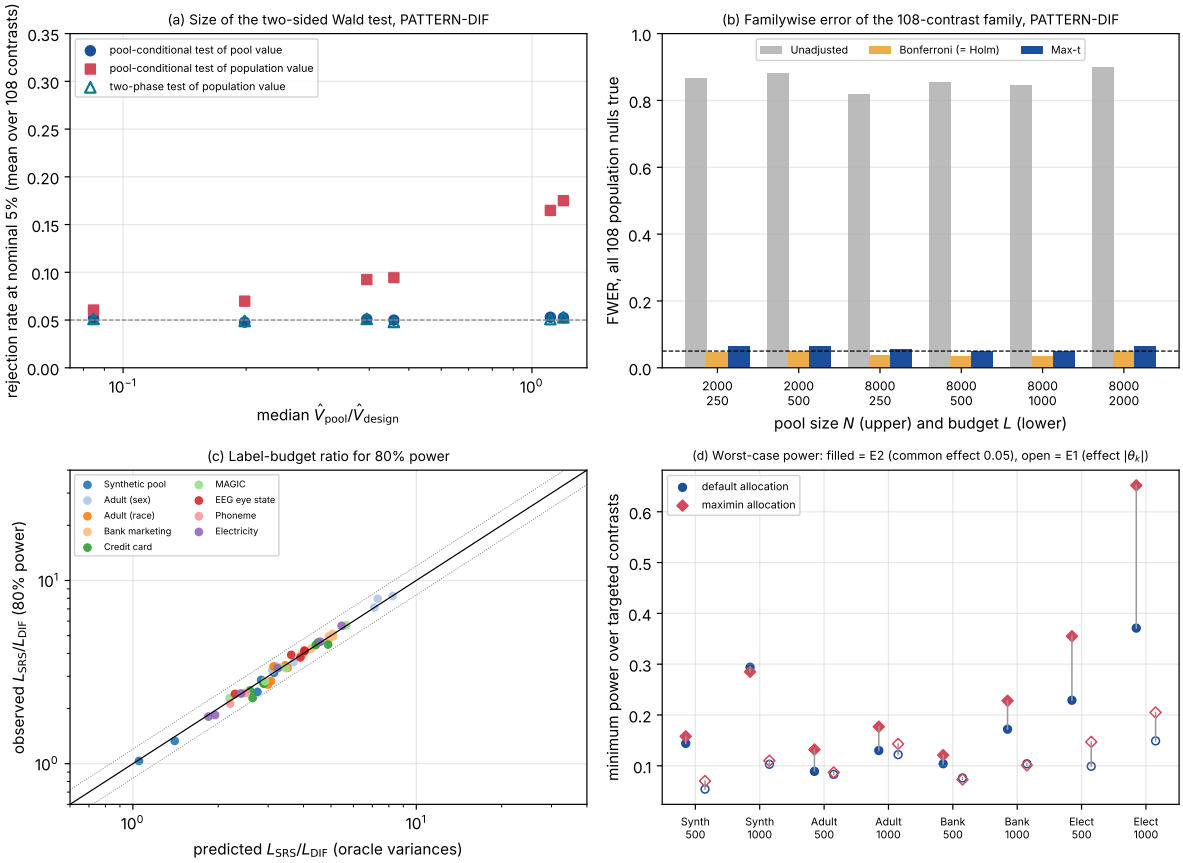


Fig. 1 (a) Rejection rate at nominal 5% (mean over 108 contrasts) against the estimated ratio of pool to design variance. (b) FWER of the family in which all 108 population hypotheses are true, by cell (N above, L below). (c) Predicted and observed $L_{\mathrm{SRS}}/L_{\mathrm{DIF}}$ at 80% power, one point per contrast; dotted lines: ±20%. (d) Minimum simulated power over the targeted contrasts under the default and the maximin allocation (filled: E2, open: E1).

Table 2 Data settings. Pool: size of the held-out pool from which labels are drawn. P_0, P_1: positives in each group. Disagreement: share of pool instances on which two of the three most accurate models disagree, averaged over the three pairs. Groups marked "arbitrary" are median splits of a numeric feature.

Setting	Group	Pool N	Positives P_0	Positives P_1	Disagreement
Adult (sex)	sex	19537	3963	712	0.053
Adult (race)	race	19537	4249	426	0.047
Bank marketing	age	18085	1753	363	0.045
Credit card	sex	12000	1148	1506	0.030
Law school	race	8320	2038	640	0.000
MAGIC	arbitrary	7608	1084	1591	0.049
EEG eye state	arbitrary	5992	1183	1506	0.086
Phoneme	arbitrary	2162	388	247	0.095
Spambase	arbitrary	1841	490	235	0.043
Churn	arbitrary	2000	149	134	0.041
Electricity	arbitrary	12000	2582	2488	0.086

does increase with the budget (a median increase of 23% from $L = 250$ to $L = 1000$), because a larger budget labels a larger share of the small disagreement strata, where the variance of the comparisons is concentrated. The increase is moderate except at $L = 2000$, where the remaining settings are those with the largest pools and the strongest gains.

Where the gain comes from.

Table 3 separates the two ingredients, and the design is the larger. Under the difference estimator, moving from simple random sampling to the PATTERN strata multiplies the efficiency by 3.3 to 8.6 (PATTERN-DIF against SRS-DIF), and under the plain estimator by 3.7 to 7.3 (PATTERN against SRS). The estimator contributes 1.3 to 1.65 on top of the PATTERN design (PATTERN-DIF against PATTERN, better in 34 of 35 cells), but only 1.8 to 2.0 on top of simple random sampling (SRS-DIF against SRS). The two ingredients are complementary, and neither accounts for the whole gain. Stratifying by group alone gives almost no gain under the plain estimator (median 1.04 to 1.05 at the first three budgets and 1.25 at $L = 2000$, where it is 1.0 without law school) and 2.1 to 2.7 with the difference estimator, which is essentially the value of the proxy correction. The variance of a contrast between two models is driven by the instances on which they disagree, and these are spread over both groups.

By estimand.

Efficiency differs by estimand. For PATTERN-DIF the medians over settings at the four budgets are 5.8, 5.3, 8.9 and 17.0 for θ^{TPR}, 7.6, 8.5, 8.4 and 20.6 for θ^{FPR}, and 3.5, 4.1, 4.4 and 5.7 for θ^{ERR}. The fairness contrasts gain more than the error contrast because they depend on the rarest cells (positives or negatives in the minority group). In Adult with race as the group, at $L = 1000$, the efficiencies are 23.9 for θ^{TPR}, 20.6 for θ^{FPR} and 3.7 for θ^{ERR}. For the pairs formed by the three most accurate models, the comparisons most likely to matter in practice, the median efficiency of PATTERN-DIF is 5.8, 6.8, 6.6 and 10.5 at the four budgets (3.3, 4.2, 3.9 and 6.2 for PATTERN).

7.3 Efficiency and disagreement

The gain is larger when the models disagree less (Figure 2). The Spearman correlation between the disagreement rate of the three best models and the efficiency of PATTERN-DIF is -0.58 at $L = 250$ (ten settings), -0.53 at $L = 500$ (nine) and -0.71 at $L = 1000$ (seven), excluding law school (-0.68, -0.66 and -0.81 with it). The three settings with the highest disagreement (EEG, electricity and phoneme, 8.6% to 9.5%) have the smallest gains, between 3.7 and 5.8 at $L = 500$. The relation is inexact, since efficiency also depends on the size of the minority cells and on the magnitude of the contrasts (Section 5; checked in Supplementary Section S5).

Two settings are atypical. In the law-school data all nine models predict almost identically (the three best models disagree on fewer than 0.01% of instances); the design then labels almost every disagreement instance and the contrasts are estimated nearly exactly, so the efficiency ratios (about 270 at $L = 500$, 510 at $L = 1000$ and above 10^4 at $L = 2000$) show only that SRS wastes labels on agreements. In the churn data (a pool of 2,000) the efficiency of PATTERN-DIF at $L = 500$ is 61, driven by contrasts of false positive rates and of errors that are concentrated in very few instances.

7.4 Inference and decisions

Table 5 reports the coverage of nominal 95% intervals. The mean coverage of the PATTERN-DIF intervals is 0.940 at $L = 250$, 0.946 at $L = 500$ and between 0.950 and 0.952 at larger budgets; for PATTERN with the plain estimator it is 0.938, 0.944, 0.950 and 0.956. Under SRS it is 0.911 to 0.942, and the worst-covered estimand has coverage as low as 0.10 at $L = 250$, because SRS samples often contain no disagreement instance in a small cell and then give a zero standard error. The worst-covered estimand under PATTERN-DIF has coverage 0.59 at $L = 250$ (in law school), 0.84 at $L = 500$ and 0.89 to 0.90 at larger budgets. The intervals are therefore adequate on average from 250 labels on, and for every estimand from about 1,000 labels on (worst-estimand coverage 0.89 to 0.90, about the minimum that chance alone gives over 108 estimands with 200 replicates). They fail in small cells at small budgets, where a stratum in which every sampled label is the same gives a variance estimate of exactly zero. A safeguard that replaces the zero variance of such a stratum by its variance under the proxy (a "variance floor", Section 9) raises the worst-estimand coverage at $L = 250$ from 0.59 to 0.82 and keeps the mean at 0.946; a more conservative version raises the mean to 0.98. The floor was designed after seeing this failure on these pools, so the improvement is in-sample. At small budgets we report both the plain Wald and the floor intervals.

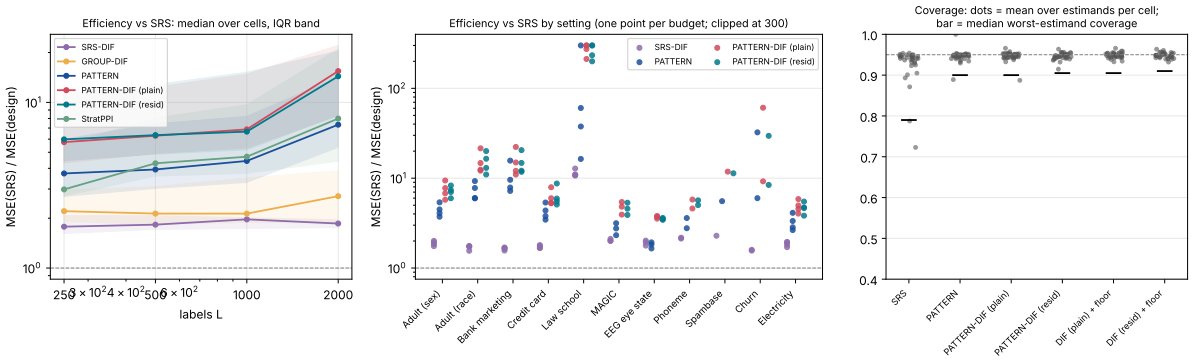


Fig. 2 Efficiency of the designs relative to SRS (median over the comparison estimands) as a function of the label budget and of the disagreement rate; see Table 4 for the numbers.

Table 3 Decomposition of the efficiency gain by budget (median over settings of the cell medians). Each row is a ratio of mean squared errors. "DIF resid vs DIF plain" compares the residual-variance allocation with the allocation used by default.

Step (MSE ratio)	$L = 250$	$L = 500$	$L = 1000$	$L = 2000$
SRS-DIF vs SRS	1.78	1.83	1.97	1.86
PATTERN vs SRS	3.72	3.94	4.43	7.33
PATTERN-DIF vs PATTERN	1.65	1.54	1.51	1.31
PATTERN-DIF vs SRS-DIF	3.28	3.29	3.72	8.57
PATTERN-DIF vs SRS	5.76	6.29	6.87	15.5
DIF resid vs DIF plain	1.01	0.97	0.96	1.00

7.5 Repeated train/pool splits

All earlier results use one train/pool split and one set of nine fitted models per setting. To measure how much the efficiency depends on the split, we repeated the main study for the six largest settings (Adult with sex and with race as the group, Bank marketing, credit-card default, MAGIC and Electricity; pools of 7608 to 19537 instances) on three new splits, with seeds 1, 2 and 3. For each split the nine models were refitted on the new 60% training part (the learners' own random seeds were not changed, so the variation comes from the data split and the refit), the pool is the new 40% held-out part, and the protocol of Section 7 was run for SRS, GROUP, PATTERN and PATTERN-DIF with $L \in \{500, 1000\}$ and $R = 100$ replicates per cell. The original split (seed 0, $R = 200$) is included as the fourth value. The code is `experiments/label_efficiency_splits.py`; the numbers are in `results/label_efficiency_splits.json` and `results/label_efficiency_splits_summary.json`. Efficiency is defined as before, the median over contrasts of the ratio of mean squared errors against SRS, computed on the pool of each split.

Table 6 and Figure 3 show that the efficiency is stable across splits relative to its variation across settings, which ranges from 4.9 (Electricity, mean over the four splits) to 15.4 (Adult with race) at $L = 1000$. The median over settings of the coefficient of variation across the four splits is 0.07 and the median ratio of the largest to the smallest split value is 1.18. The largest spreads are Adult (race) at $L = 500$ (from 10.6 to 14.6) and Credit card at $L = 1000$ (6.0 to 7.9). Part of this spread is Monte Carlo error, since each value rests on 100 replicates (200 for split 0). Adult (race) and Bank have the highest efficiency and Electricity and MAGIC the lowest in every split and at both budgets. The efficiency of PATTERN-DIF is at least 4.2 in all 48 cell-split values.

The median over the six settings is 6.0, 6.7, 6.5 and 7.4 at $L = 500$ for splits 0 to 3, and 6.9, 6.9, 7.6 and 7.3 at $L = 1000$. The original split gives the lowest median of the four at both budgets (at $L = 1000$ only marginally below split 1), so the choice of split does not flatter the headline

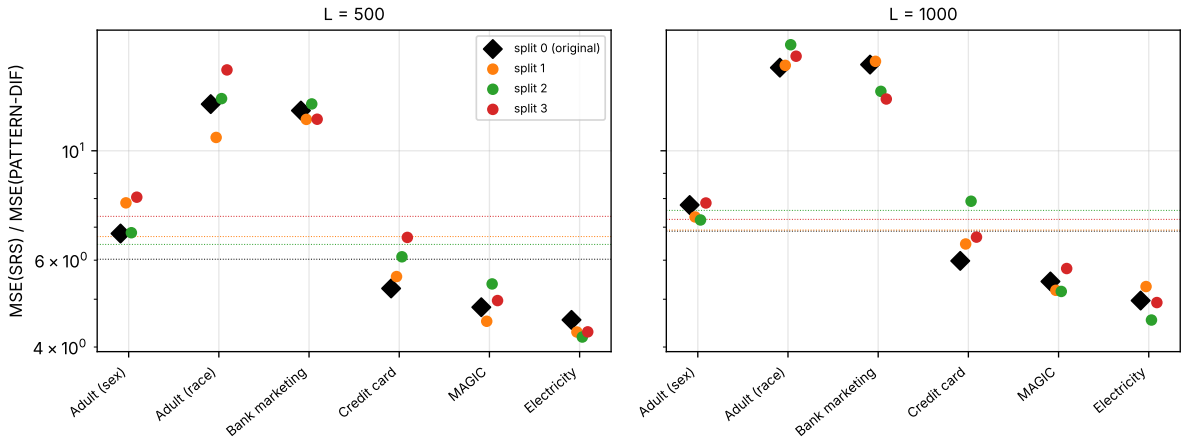


Fig. 3 Efficiency of PATTERN-DIF against SRS by setting and split. The diamond is the original split. Dotted lines are the medians over the six settings for each split.

numbers of the main study in these settings. Pooling the four splits, the median is 6.7 at $L = 500$ and 7.3 at $L = 1000$ against 6.0 and 6.9 for split 0 alone, a difference of at most about 12% on a factor of 6 to 7. Coverage is unaffected: the mean coverage of the PATTERN-DIF intervals is 0.948 to 0.951 on every split (0.950 overall), against 0.939 to 0.941 for SRS with the plug-in estimator. The worst contrast in any split has coverage 0.84 to 0.88 for PATTERN-DIF; with 100 replicates per cell the Monte Carlo standard error of one coverage is 0.02, so these worst values are partly noise.

These results have three limits. Four splits in total (three new) are far fewer than the 20 that a full study of between-split variation would use, so the spread in Table 6 is a rough indication only. Only the six largest settings were repeated, so the median over all eleven settings of the main study is not recomputed. The nine models share the same families in every split, and the splits do not change the learner seeds or the groups, so variation from other hyperparameters or model families is not covered.

7.6 Target of inference: the pool or the population

The intervals of Section 4 cover the contrast computed on the pool, $\theta_U = f(T_U)$. If the pool is itself a sample of N units from a deployment population, the quantity of interest is the population contrast $\theta_{\text{pop}} = \tilde{f}(\tau)$, $\tau = \mathrm{E}\, t_i(Y_i)$, and the pool-conditional interval ignores the pool's sampling error. Write $\hat{\theta} = f(T_{\text{dif}})$ for the unchanged estimator. Then

$$\hat{\theta} - \theta_{\text{pop}} = \underbrace{f(T_{\text{dif}}) - f(T_U)}_{\text{design error}} + \underbrace{f(T_U) - f(N\tau)}_{\text{pool error}}.$$

Given the pool, the first term has mean zero to first order and variance $V_{\text{design}} = \sum_h N_h^2(1 - n_h/N_h)S_h^2(e)/n_h$, the quantity estimated in Section 4. The second term is, to first order, a sum of N independent influence values, so the two terms are uncorrelated and

$$V_{\text{total}} = \mathrm{E}\, V_{\text{design}} + V_{\text{pool}},$$
$$V_{\text{pool}} = N \operatorname{Var}(z_i), \qquad z_i = \mathbf{b}^\top \big(t_i(Y_i) - \tau\big), \tag{30}$$

with $\mathbf{b} = \nabla f(N\tau)$ the gradient on the scale of totals. This is the two-phase decomposition of Särndal and Swensson [86]. The influence value z_i is that of the full-label plug-in contrast, not the residual e_i of the difference estimator: the proxy removes variance from the labelled part of the error but cannot remove the variation of Y across units of the population, so, to first order, V_{pool} is the same for every labelling design and every proxy.

Since the labelled sample is a stratified sample of the pool, V_{pool} is estimated with the weights

$w_i = N_h/n_h$:

$$\hat{V}_{\text{pool}} = \sum_{i \in s} w_i \, (\hat{z}_i - \bar{z})^2,$$

$$\bar{z} = \frac{1}{N} \sum_{i \in s} w_i \hat{z}_i, \qquad \hat{z}_i = \hat{\mathbf{b}}^\top t_i(Y_i),$$

with $\hat{\mathbf{b}} = \nabla f(T_{\text{dif}})$, and the two-phase interval is $\hat{\theta} \pm 1.96\sqrt{\hat{V}_{\text{design}} + \hat{V}_{\text{pool}}}$. The derivation is in `docs/two_phase_note.md` and the code in `tcs/labeling_two_phase.py`. Let $\varpi = V_{\text{pool}}/V_{\text{design}}$. It is of order $L\,\text{Eff}/N$, where Eff is the efficiency of the design against SRS, so it grows with the budget and the efficiency and shrinks with the pool size. With correct V_{design} and normal errors the pool-conditional interval has coverage about $2\Phi(1.96/\sqrt{1+\varpi}) - 1$ of the population contrast, which is 0.93 at $\varpi = 0.2$, 0.83 at $\varpi = 1$ and 0.67 at $\varpi = 3$. Efficiency changes in the same way: the pool term is shared by SRS and PATTERN-DIF, so against the population contrast the MSE ratio is $(\text{Eff} + \varpi)/(1 + \varpi)$, with ϖ evaluated for PATTERN-DIF, rather than Eff. For Eff $= 6$ and $\varpi = 1.5$ this is 3. This follows arithmetically from (30); it is not an estimate from the earlier experiments.

Experiment

We checked the two intervals against population values in a synthetic setting; the real data sets are not used. The population is the generative model of the non-smooth simulation (five Gaussian covariates, a group with prevalence 0.3, a logistic label model) with nine fixed scoring rules, so there are 36 pairs and 108 contrasts, θ^{TPR}, θ^{FPR} and θ^{ERR}. Population values were computed on 6×10^6 draws. For every replicate a new pool of $N \in \{2000, 8000, 20000\}$ units is drawn and then a PATTERN-DIF sample (PATTERN strata, plain allocation, difference estimator) of $L \in \{250, 500, 1000, 2000\}$ labels ($L \le N/4$; ten cells), with the mean-probability proxy; $R = 300$ replicates per cell. Table 7 and Figure 4 give the coverage, averaged over the 108 contrasts and for the worst contrast, together with the median over contrasts of $\hat{V}_{\text{pool}}/\hat{V}_{\text{design}}$ (source: `results/two_phase_check.json`).

The pool-conditional interval keeps its coverage of the pool value (mean 0.945 to 0.953 in all ten cells), so the earlier results are not affected. Against the population contrast its mean coverage falls with ϖ: 0.94 when $\varpi \le 0.11$, 0.90 to 0.93 for ϖ between 0.2 and 0.5, 0.89 at $\varpi = 0.58$ and 0.79 to 0.81 at $\varpi \approx 1.5$ (the cells $N = 2000, L = 500$ and $N = 8000, L = 2000$), where the worst contrast is covered in 0.72 and 0.64 of the replicates. This is close to the value the formula predicts. The two-phase interval has mean coverage 0.945 to 0.952 in every cell and worst-contrast coverage 0.90 to 0.92. The worst of 108 contrasts with 300 replicates has a Monte Carlo standard error of 0.013, so the worst value is expected to lie near 0.92 even for exact intervals. We do not read the remaining gap as a failure, but the experiment cannot exclude a small one. The cost is width: the two-phase interval is 2% to 10% wider when $\varpi \le 0.2$, 12% wider at $\varpi = 0.25$ and 21% to 71% wider when $\varpi \ge 0.44$ (median width over contrasts).

When the pool term matters

It does not matter when N is large relative to the budget: at $N = 20000$ and $L \le 500$, $\varpi \le 0.09$ and the two intervals are nearly the same. It matters when the labelled fraction is not small, as in the small pools of the empirical study (about 2000 to 20000 instances, budgets up to 2000). The empirical MSE of $\hat{\theta}$ around the population contrast is 1.04 to 2.6 times its MSE around the pool contrast in these cells. Three limits apply. The simulation is stylised: the pool is an independent sample of the population and the models are fixed and independent of the pool. With real data the deployment population is unknown, the pool may be a convenience sample, and the two-phase interval is only as good as the assumption that the pool is representative. If the pool is the whole deployment set, the pool-conditional interval is the right one and the two-phase interval is conservative. The gain of PATTERN-DIF over SRS also shrinks for the population target, as shown above, although the ranking of designs does not change. We did not run the full pipeline (training, pooling, labelling) against full-data references on the OpenML data; the known population above isolates the pool-sampling term.

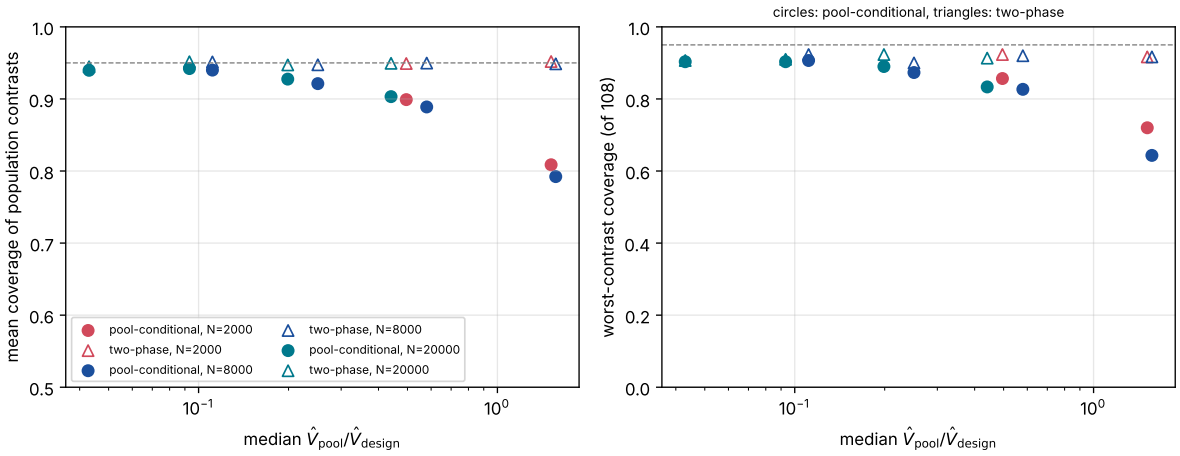


Fig. 4 Coverage of the population contrasts against the median ratio $\hat{V}_{\text{pool}}/\hat{V}_{\text{design}}$, one point per (N, L) cell. Left: mean over the 108 contrasts. Right: worst contrast. Circles: pool-conditional intervals, triangles: two-phase intervals; colours give N.

8 Comparison with existing labelling designs

PATTERN-DIF combines four ingredients that are each known: sampling where models disagree, stratification on a model-based proxy with Neyman-type allocation, design-based ratio estimation and a model-assisted (difference) estimator. This section asks whether natural alternatives built from one or two of these ingredients already do as well. The study was run before the final form of the method was fixed. Its PATTERN-DIF variant (B5, $\lambda = 1$) is the same estimator as the main method, but the samples differ from those of Section 7 (different random seeds). Values differ between the two studies by a median of about 5% per cell (up to 27% in single cells, law school excluded) and the medians over settings by up to 14% (PATTERN-DIF at $L = 250$: 6.6 here, 5.8 in Section 7). The abstract and the main results use the study of Section 7. We ran the baselines and SRS and PATTERN on the eleven settings of Section 7, at $L \in \{250, 500, 1000, 2000\}$ with $L \leq N/4$ (35 setting–budget cells) and $R = 200$ replications per cell. To isolate the effect of labelling and estimation, every method estimates the same totals T, maps them through the same $\hat{\theta} = f(\hat{T})$ to the same 108 comparison estimands, uses the same proxy π_i and allocation weights where it has any, and reports 95% Wald intervals; the budget is counted in distinct labelled instances. Estimands that every design estimates exactly are excluded (51 of 108 in Law school, none elsewhere). Full definitions are in Supplementary Section S6.

Baselines.

B1 is importance sampling for active comparison [5]: draws with replacement from an instrumental distribution that is proportional to the root of the proxy-expected squared influence, adapted here to TPR and FPR gaps, with self-normalised totals. B2 stratifies on the proxy score (group × quantile bins of π_i; Score-10 and Score-5) with the same proxy-Neyman allocation as PATTERN [13, 14], so it differs from PATTERN only in the strata. B3 is a disagreement-focused two-phase design (Dis-25, Dis-50): a simple random first phase of $L/4$ or $L/2$ labels, then a simple random sample of the disagreement set for the rest [5, 8]. B4 is prediction-powered inference: PPI with $\lambda = 1$ [16], PPI++ with a tuned λ [57] and StratPPI, which applies PPI++ within proxy strata with a Neyman-type allocation [15]. B5 is the proposed method: the PATTERN design with the stratified difference estimator ($\lambda = 1$, PATTERN-DIF), and a tuned-weight variant (PATTERN-PPI).

Results.

Table 8 summarises all methods by budget. PATTERN is the most efficient of the designs that use the same plain estimator as SRS (B1

to B3) at every budget: its median efficiency against SRS is 3.75, 3.95, 4.35 and 7.43 at $L = 250, 500, 1000, 2000$. It is not the most efficient method overall: StratPPI, which combines proxy strata with a model-assisted estimator, matches or exceeds it in the paired medians at $L = 250$, 500 and 1000 (ratios 1.01, 1.12, 1.08) and has the higher median efficiency against SRS at $L =$ 500, 1000 and 2000 (4.45, 4.58, 8.50), and the extensions B5, which add the same kind of correction to the PATTERN sample, beat both. Among the competing designs the closest are importance sampling (B1: 3.09, 2.77, 3.48, 4.30 against SRS; 0.54–0.79 of PATTERN) and StratPPI (B4: 3.07, 4.45, 4.58, 8.50; 1.01, 1.12, 1.08, 0.83 of PATTERN). Proxy-score strata alone (B2) reach 1.45–4.16 against SRS, the disagreement-only designs (B3) 1.7–2.5 and PPI/PPI++ with simple random labels 1.8–2.1, all clearly below PATTERN (medians 0.25–0.57 of its efficiency). PPI and PPI++ with SRS labels barely move with L: they use the proxy but not the fact that the numerators vanish where models agree. The model-assisted extensions improve on PATTERN in every cell except Law school at $L = 2000$, where PATTERN is already exact, by a median factor of 1.3–1.7 depending on budget and variant and up to 11 in Law school at $L \leq 500$; PATTERN-DIF reaches 6.6, 6.2, 6.6 and 15.5 against SRS. The estimated-λ version adds little (within 3% of PATTERN-DIF in the median) and costs coverage, so we recommend the fixed-λ version. The extension changes the estimator and leaves the design alone. Its design-unbiasedness and variance estimator are covered by the theory of Section 5, which treats the difference estimator with a fixed coefficient; the tuned-coefficient version is covered only by the asymptotic equivalence statement there.

Where a baseline beats PATTERN.

Supplementary Table S15 and Supplementary Figure S5 (Supplementary Section S6) show per-setting ratios. First, StratPPI beats PATTERN at $L = 500$ in seven of the ten settings of the table (ratios 1.01–1.57) and is below it in Adult-sex (0.97), Adult-race (0.93) and Churn (0.24); over all cells its ratio exceeds 1 in 19 of 35. The margin is small in most settings and reflects the model-assisted estimator that StratPPI uses: the same correction applied to the PATTERN sample improves PATTERN itself by a median factor of 1.3–1.7. Second, Dis-25 beats PATTERN on Churn at $L = 500$ (ratio 1.52), where the disagreement set is only 19% of the pool, but its intervals are unreliable there (mean coverage 0.93, worst-estimand coverage 0.66). Third, PPI++ beats PATTERN on EEG (ratio 1.19 at $L = 500$), the setting with the largest disagreement set (73%). Importance sampling never beats PATTERN in the all-estimand median (ratio below 1 in all 35 cells), but on the pairs among the three most accurate models it is competitive: its median efficiency against SRS on those pairs is 4.2, 4.1, 4.2 and 6.3 against 3.4, 3.6, 4.0 and 6.8 for PATTERN, and it wins on those pairs in 10 of 35 cells. If only the best few models matter, importance sampling is a serious alternative. Conversely, in the two settings where PATTERN gains most (Law school and Churn, whose disagreement sets cover only 19% of the pool) it is 6–84 times more efficient than SRS at $L \leq 1000$ and labels essentially every disagreement instance at $L = 2000$ in Law school, because it gives the agreement region a stratum of its own that can be labelled sparsely.

Coverage.

Over all 35 cells the mean 95% coverage is 0.925 (SRS), 0.947 (PATTERN), 0.950 (IS), 0.941 (Score-10), 0.942 (Dis-25), 0.929 (PPI++), 0.924 (StratPPI), 0.946 (PATTERN-DIF) and 0.929 (PATTERN-PPI). Shortfalls concentrate in the smallest cells. The lowest worst-estimand coverages are 0.14 for StratPPI (Churn, $L = 250$), 0.16 for SRS and PPI++, 0.20 for PATTERN-PPI and 0.52 for PATTERN. The tuned estimators undercover when per-stratum weights are estimated from a few labels; PATTERN-DIF, which fixes $\lambda = 1$, keeps the coverage of PATTERN. Undefined estimates occur in at most 1% of replications for any method. Details are in Supplementary Section S6.

Limitations.

The comparison is restricted to fixed-budget, one-shot designs on cached pools of nine models. The proxy is the same for every method, so a better proxy could change the ranking, especially for StratPPI and PATTERN-PPI, which rely on it directly. The importance-sampling baseline draws

with replacement and so spends a median of $1.12L$ draws for L labels; counting draws instead of labels would lower its efficiency. With $R = 200$ replications the Monte Carlo error of a single mean squared error is about 10%, so ratios between 0.9 and 1.1 should not be read as a ranking, and medians over estimands hide differences between estimands.

9 Ablations and robustness

This section varies one ingredient of PATTERN-DIF at a time and asks whether the estimator remains valid and how much efficiency is gained or lost relative to the default. The default is the method of the main study: strata (G_i, K_i) with strata smaller than 50 merged within group, weighted generalised Neyman allocation [65] computed with the ensemble-mean proxy π_i, a floor of 10 labels per stratum, the stratified difference estimator with $\lambda = 1$ and the same proxy, and 95% Wald intervals from the linearisation variance. The plain (Horvitz–Thompson) estimator, which uses the proxy only to allocate labels, is called HT, $\lambda = 0$ or "plug-in" in the tables. The ablations are numbered A1 to A5 as in the Supplementary Information (A1 stratifier, A2 proxy and weight λ, A3 number of models, A4 small-sample inference, A5 ranking by the equalised-odds gap). The proxy enters both the allocation and the estimator, so the proxy ablations report the two roles separately. Full details are in Supplementary Section S7.

Protocol.

We use the eleven cached pools and budgets $L \in \{250, 500, 1000\}$ with $L \leq N/4$ (29 setting–budget cells), $R = 150$ replications per cell and design, and the 108 comparison targets of the 36 pairs of nine models (targets that are exact for every design are excluded). Efficiency is the per-target ratio $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{design}}$ with SRS and the plain estimator as reference, summarised by the median over targets and then over cells. "Rel." is the median over cells of $\mathrm{MSE}_{\text{PATTERN-DIF}}/\mathrm{MSE}_{\mathrm{variant}}$ (above one favours the variant). With $R = 150$ the minimum over 108 targets of coverage that is exactly nominal has mean 0.90, so worst-target coverage near 0.90 is consistent with nominal behaviour, and values well below 0.88 are not.

Stratifier.

On the same samples the difference estimator is more efficient than HT in all 29 cells (median ratio 1.67), and its efficiency against SRS is 6.3 at every budget, compared with 3.6 to 4.3 for HT. The gains are not additive: SRS with the difference estimator gains a factor 1.9, PATTERN with HT 4.0, and the product (7.5) exceeds the combined gain. The count K_i is the useful part of the stratifier (Table 9): group × agree/disagree keeps 74% of the default's precision, the joint pattern of the three models agreeing most with the ensemble keeps 38%, and splitting (G, K) cells by proxy bins is a tie (Rel. 0.99 to 1.01), whose extra strata eat the budget through the floor. A residual-variance allocation is not better (Rel. 0.96). The references that drop an ingredient lose clearly: GROUP-DIF and SRS-DIF keep 28% and proportional allocation 57%. Coverage depends little on the stratifier (mean 0.938 to 0.946).

Proxy.

The allocation is insensitive to the proxy; the estimator is sensitive to it. With HT, every allocation proxy gives a median efficiency of 3.95 to 4.05 against 3.97 for the ensemble mean, including the reversed proxy $1-\pi$, a shuffled proxy and the constant 0.5, because the influence values vary within a stratum mainly through the differences $\hat{y}_{mi} - \hat{y}_{ji}$, which need no labels; the allocation matters only in being Neyman-type and not proportional (Rel. 0.57). With the ensemble proxy kept in the estimator, alternative allocation proxies keep 99% to 101% of the default's precision. When the proxy is also used in the estimator, replacing the ensemble mean by one logistic regression costs 13%, naive Bayes 42%, the constant 16%, a shuffled proxy 47%, a noisy proxy 39% and the reversed proxy 34%. The difference estimator stays unbiased and its coverage is unaffected (mean 0.940 to 0.949), so a poor proxy costs efficiency only, although a corrupted proxy also removes the gain over HT (median DIF/HT ratio 0.96 for shuffled and noisy proxies against 1.67 for the ensemble). The per-proxy table and figure are in Supplementary Section S7 (Supplementary Table S16).

The weight λ.

A λ tuned per stratum and target and clipped to $[0, 1]$ [as in PPI++ and StratPPI; 15, 57] is slightly

more efficient than $\lambda = 1$ with the ensemble proxy (median Rel. 1.016) but undercovers: mean coverage 0.923 and worst target 0.799 on average (0.227 in Law school at $L = 250$), against 0.945, 0.877 and 0.633 for $\lambda = 1$. The weights are estimated from a few labels per stratum and the variance estimate ignores that. No alternative beats $\lambda = 1$ in both efficiency and validity with a good proxy, so we keep $\lambda = 1$ (Supplementary Table S27); tuning is worth considering only when the proxy is known to be poor and some undercoverage is acceptable.

Number of models.

For $M \in \{2, 3, 5\}$ (random subsets of the nine models) the gain depends on the share of the pool on which the compared models disagree more than on M: the Spearman correlation between log efficiency and disagreement share is -0.72, -0.74 and -0.76 at $L = 250$, 500 and 1000. At $L = 250$ the median efficiency of PATTERN-DIF is 17.2 when the models disagree on less than 10% of the pool, 12.5 for 10 to 20%, 4.8 for 20 to 40% and 3.8 above 40%. The gain of DIF over HT grows with M, from 1.15 at $M = 2$ to 1.67 at $M = 9$. Coverage does not depend on M (mean 0.945 to 0.950); see Supplementary Table S17 and Supplementary Figure S6.

Small-sample inference.

With the plain estimator, worst-target coverage is near 0.5 at $L = 250$ in Churn and Law school because some strata show no variation in the influence values. The difference estimator removes most of this. Its residual $(Y_i - \pi_i)(u_i - v_i)$ varies even when no positive is sampled, no (replication, target) pair has a standard error of exactly zero, and the worst-target Wald coverage in Churn at $L = 250$ rises from 0.47 to 0.91 (Table 10). Shortfalls remain: the worst-target Wald coverage is below 0.85 in five of 29 cells (Law school 0.63, MAGIC 0.78 and Adult-sex 0.82 at $L = 250$, Bank 0.83 and Phoneme 0.83 at $L = 500$). The mean standard error is 0.90 of the Monte Carlo standard deviation and the bias is small, so the intervals are too short and not off-centre, consistent with a linearisation error in ratios with few sampled positives. A Satterthwaite t quantile adds 1% to the width and barely helps, and the stratified bootstrap is 11% wider with worst target 0.878 and minimum 0.767. The proxy variance floor replaces a stratum variance that is exactly zero by its proxy residual variance ('zero') or takes the larger of the two in every stratum ('max'). The 'zero' floor leaves the width unchanged and lifts the minimum over cells from 0.633 to 0.780, entirely through Law school; only the 'max' floor reaches 0.92 in every cell, at 1.25 times the Wald width in the median (4.9 in Law school). The floors were designed after seeing the failure of the plain estimator on these pools, so the 'zero' result for Law school is in-sample; a check on synthetic pools (`results/floor_check.json`; Supplementary Section S5) finds that the floor leaves the interval width unchanged (median ratio 1.00), but no zero standard error occurs there, so it does not test whether the floor repairs coverage.

Ranking models by the equalised-odds gap.

Practitioners also rank models by their equalised-odds gap $EO_m = \max(|\Delta_m^{TPR}|, |\Delta_m^{FPR}|)$ [1]. The plug-in estimate is biased upwards, as any maximum of absolute values is. PATTERN-DIF is more accurate than SRS in the median (efficiency 1.94 to 2.21) and identifies the best model more often (mean probability 0.50 against 0.35 at $L = 250$), but the difference estimator adds little over HT for this purpose, and PATTERN-DIF is less accurate than SRS in all Electricity cells (0.57 to 0.73), because the allocation spends labels on disagreement, whereas the gap of one model depends on all its positive predictions. In the three Law school cells, where the models are almost identical, the difference estimator is far worse than SRS for this purpose (efficiency about 10^{-4}; best model identified in 0.45 to 0.59 of the replicates against 1.00), whereas the plain estimator on the same design is far better (38 to 208); we have no tested explanation. A variant whose allocation targets the $2M$ per-model gaps is the best design here (efficiency 2.30 to 2.61) but keeps only 74% of the precision for the comparison contrasts. When the true margin to the next model is 0.002 or less, identification is hard for every design (Supplementary Table S18).

Recommendations.

The count K_i is the stratifier to use, Neyman allocation beats proportional allocation, the allocation is insensitive to the proxy, and the gain depends on how often the compared models disagree. The proxy matters in the estimator, where a poor proxy costs up to half of the efficiency and removes the advantage over HT, although coverage is unaffected; no alternative proxy beat the ensemble mean in the median. Neither tuned λ nor proxy-bin strata beats the default consistently, so we keep $\lambda = 1$ and K_i strata. We recommend reporting the Wald interval with the 'zero' floor, which costs nothing in width; the remaining undercoverage of rate contrasts at $L = 250$ is not fixed by t quantiles or the bootstrap, and only the conservative 'max' floor reaches nominal worst-target coverage. For ranking models by EO_m itself, the gap-targeted allocation is better than the default outside Law school, where the difference estimator failed for this purpose.

10 Practical guidance

This section gives a procedure for comparing a small set of candidate classifiers on fairness and accuracy with a limited labelling budget.

Step 1: describe the agreement structure before labelling.

Compute K_i and the stratum sizes N_h on the unlabeled pool, and the pairwise disagreement rates of the models to be compared. These numbers cost no labels and indicate the benefit of the design (rank correlation -0.58 to -0.71 with the efficiency at $L = 250$ to 1000, law school excluded, Section 7): in our data, disagreement rates below 5% went with efficiencies of 5.4 to 15.0 at $L = 1000$ (law school excluded), while rates of 8.6% went with efficiencies of 3.8 to 5.0. If the candidate models disagree on a large share of the pool, the gain is modest.

Step 2: check that the minority cells are not empty.

Every group-wise rate needs positive and negative instances in each group. The proxy probabilities give an expected count of positives per group and stratum before any label is collected; if the expected number of minority positives in the pool is very small, no design can estimate the corresponding rates, and the estimand should be reconsidered (for example by merging groups).

Step 3: choose the estimands and allocate.

Use the signed contrasts (4) for the pairs of interest. If only a few pairs matter, restricting the estimand set to them concentrates the labels further. Compute the allocation (9) with the average predicted probability as proxy, at least ten labels per stratum, draw the instances in each stratum without replacement and label them.

Step 4: estimate and report uncertainty.

Estimate the totals with the difference estimator (7), which corrects the pool-wide proxy total by the labelled residuals and needs no tuning, and report the contrasts with standard errors and intervals. If the statement is about the deployment population rather than about the pool itself, add the pool-sampling variance (Section 7.6): the pool-conditional intervals undercover once the labelling budget is a sizeable fraction of the pool, with coverage of the population contrast down to 0.79 when the variance ratio exceeds one, while the two-phase intervals keep it near 0.95. For budgets below about 500 labels or pools below a few thousand instances, use the variance floor of Section 9 (and, if worst-contrast coverage matters, its conservative version, which widens the intervals by about a quarter in the median) and report the stratum sample sizes, so that readers can judge how many minority positives informed each rate.

Step 5: test and decide.

State the hypotheses before labelling: equality of the gaps of two models, or non-inferiority of a candidate against an incumbent with a margin δ chosen for the application. For a single prespecified contrast use the Wald test of Section 6. When many contrasts are examined, use the Holm or the max-t procedure, which control the familywise error rate asymptotically; in our simulations max-t had familywise error slightly above nominal (0.058 at the 5% level) and, on real pools, simultaneous coverage below that of Bonferroni at the smallest budgets (Section 6.7), so Holm is the safer default when the budget is small. A contrast whose test rejects supports a statement that one model has a smaller gap than the other on

the stated metric. When none rejects, the budget has not resolved the comparison. To plan the budget, use the sample-size formula of Section 6: for a fixed allocation the labels needed for a given power are inversely proportional to the squared effect, and the design effect of Section 7 converts the number of simple random labels into the number needed here; if some contrasts matter more, the power-optimal allocation of Section 6 favours them.

When not to use the design.

The design compares fixed models. It does not estimate a single model's fairness more precisely than a group-stratified design does, because one model's group-wise rates do not depend on disagreements with other models. It also assumes that the group attribute is observed for the whole pool. When the attribute is missing for some instances, it can be added as a third stratum level, but the fairness estimands then refer only to instances with a recorded group.

11 Discussion

Summary.

Comparing classifiers on group-fairness criteria is expensive in labels because the relevant information sits in small cells: the positives and negatives of the minority group, and the instances on which the models disagree. PATTERN-DIF stratifies the pool by group and by the number of models predicting the positive class, allocates labels with a proxy-based generalised Neyman rule, and estimates the contrasts with a difference estimator that corrects the pool-wide proxy total by the labelled residuals. Across eleven data settings with nine models it reduces the labels needed for a given accuracy by a median factor of 5.8 at 250 labels, 6.3 at 500, 6.9 at 1,000 and 15.5 at 2,000 (5.5, 5.8, 6.0 and 9.4 without the degenerate law-school setting). The stratification contributes more than the estimator (a factor of 3.3 to 8.6 against 1.3 to 1.65), and the two are complementary. Stratifying by group alone gives no gain with the plain estimator. On the inferential side, the Wald and simultaneous tests have asymptotic size α, their power function gives the labels needed for a target power (which shrink by the efficiency factor, to within about 20% in simulation), and familywise error is controlled over all contrasts. The design needs no label to set up and keeps the design-based guarantees of stratified sampling. Its benefit tends to be larger when the candidate models disagree less, which can be checked before any label is collected, although the rank correlation is only about -0.6 and the forecast is rough.

What the results do and do not show.

The efficiency figures are ratios of mean squared errors against simple random sampling on pools with known labels, so they give the number of labels the design saves for the same accuracy on the same pool. Eleven data sets are not representative of all comparisons an analyst will face. The gain depends on the disagreement of the compared models: it is between 3.7 and 5.8 at 500 labels for the settings with the highest disagreement (8.6% to 9.5%), and larger where the models are nearly identical. The estimator helps because the average predicted probability of nine models is informative in these data (area under the ROC curve 0.76 to 1.00) and, outside EEG, only moderately miscalibrated (10-bin calibration error 0.02 to 0.07; 0.20 in EEG). An uninformative or inverted proxy removes the advantage of the difference estimator over the plain one (Section 9), and the theory shows that the difference estimator loses whenever the within-stratum regression slope of the true influence on its proxy falls below one half (Section 5). The allocation is insensitive to the proxy.

Relation to earlier work.

Sampling where models disagree is the central idea of active comparison of prediction models [5], and verification designs that oversample discordant results have compared diagnostic tests in medicine for decades. Model-assisted estimation with predictions, including stratified versions, is the subject of prediction-powered inference [15, 16]. A stratified prediction-powered estimator that stratifies by the proxy score does not beat PATTERN-DIF in the paired median ratio of mean squared errors in any of the 35 cells of the main study (0.5 to 0.6 by budget, Section 7); its median efficiency against SRS exceeds that of PATTERN-DIF in one cell, Bank at 2,000 labels. Three features are specific to the present setting: the estimands are ratios whose numerators depend

on disagreements and whose denominators depend on group prevalence everywhere; many contrasts across several models must be estimated from one sample; and the group is known before labelling. The stratification by the count of positive predictions, the allocation rule and the difference estimator together handle all three.

Limitations.

First, the main experiments use one split (three more for six settings), one set of nine models and pools of 1,841 to 19,537 instances per setting; larger and more diverse collections of models would change the disagreement structure and with it the gain. Second, in six of the eleven settings the group is arbitrary, so these settings test the statistics of the design and carry no fairness meaning. Third, the intervals are asymptotic. Mean coverage is 0.94 to 0.95, but at 250 labels the worst-covered contrast, in the law-school setting, has coverage 0.59. A variance floor for strata whose sampled labels are all equal raises the lowest value over settings from 0.59 to 0.82 (main study), and only a conservative version reaches the nominal level, with intervals 25% wider in the median and 67% wider on average (ablation study, Section 9). The floor was designed after seeing these failures, so its benefit is in-sample. Fourth, the primary estimands are signed gaps; the equalised-odds gap is a non-smooth function of them, and the design is a compromise for ranking models by it; the difference estimator fails for this purpose in the law-school setting (Section 9). Fifth, the testing theory is asymptotic with a fixed number of strata and contrasts and rests on high-level conditions (a negligible variance floor and a Lindeberg-type condition for linear combinations); in simulation the max-t procedure had familywise error slightly above nominal and, on real pools, simultaneous coverage below that of Bonferroni at the smallest budgets, and the power-optimal allocation needs effect sizes that are unknown in practice (Section 6.7). Sixth, inference is about the fixed models on the given pool; variation from retraining is not addressed, and the two-phase interval for the deployment population was checked only on a synthetic population. Finally, labels are assumed correct; label noise that depends on the group would bias every design, including this one.

Extensions.

The design should extend to more than two groups (one stratum dimension per group level), to other ratio metrics such as positive predictive value or calibration within groups, and to sequential labelling, in which the allocation is updated as labels arrive by replacing the proxy moments with estimated ones. A stronger proxy, for instance a calibrated or learned model of the label given the predictions, could increase the gain of the difference estimator, which grows with the share of the label variance that the proxy explains (untested here). Combining the design with a retraining-aware bootstrap would address the variation due to model fitting.

Statements and Declarations

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Competing interests.

[To be completed by the author: any financial or non-financial interests related to this work, or a statement that there are none.]

Data availability.

All data sets are public and were obtained from OpenML (Adult, Bank Marketing, default of credit card clients, law school admission, MAGIC gamma telescope, EEG eye state, Phoneme, Spambase, Churn and Electricity); versions and the construction of the settings are given in Supplementary Section S9.

Code availability.

All code, simulation outputs and the LaTeX sources are available at https://github.com/vinhqdang/Special-Issue-on-Statistics-for-Data-Science-and-AI.

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Table 4 Efficiency (ratio of mean squared errors, median over estimands) by setting and budget. DIF plain: PATTERN-DIF with the default allocation; DIF resid: with the residual-variance allocation; StratPPI: stratified prediction-powered inference (Section 8). "vs PATTERN": PATTERN-DIF relative to the plain estimator under the same design. The last three columns are the efficiencies of PATTERN-DIF by estimand type. Bold marks the best of the two PATTERN-DIF variants.

Setting	L	Efficiency vs SRS						vs PATTERN	PATTERN-DIF (plain) by estimand		
		SRS-DIF	GROUP-DIF	PATTERN	DIF plain	DIF resid	StratPPI	DIF plain	θ^{TPR}	θ^{FPR}	θ^{ERR}
Adult (sex)	250	1.75	2.20	3.72	5.76	**5.98**	3.28	1.37	8.68	7.59	3.50
	500	1.86	1.95	4.08	6.80	**7.02**	4.07	1.49	9.30	8.20	3.83
	1000	2.01	1.91	4.49	**7.76**	7.39	4.71	1.51	11.8	8.75	4.28
	2000	1.94	1.87	5.39	**9.42**	8.30	5.54	1.32	11.7	11.8	5.03
Adult (race)	250	1.56	3.31	5.96	**12.1**	11.0	5.46	1.65	18.1	15.4	2.96
	500	1.73	3.33	6.02	12.4	**13.0**	6.95	1.44	17.5	14.9	3.44
	1000	1.76	3.36	7.78	14.7	**16.5**	8.52	1.48	23.9	20.6	3.70
	2000	1.75	3.52	9.28	**21.5**	20.0	10.4	1.30	37.5	29.3	4.36
Bank marketing	250	1.57	3.84	7.22	11.0	**11.8**	7.80	1.44	11.0	18.6	5.05
	500	1.64	3.99	7.87	12.1	**12.1**	10.3	1.45	12.1	24.7	6.31
	1000	1.67	3.90	9.60	**15.0**	14.8	13.1	1.43	15.6	38.9	6.56
	2000	1.69	3.97	15.7	22.3	20.5	**24.5**	1.32	22.3	79.9	9.18
Credit card	250	1.67	1.86	3.45	**5.31**	5.09	2.58	1.26	3.55	7.18	4.27
	500	1.80	1.81	3.79	5.26	**5.53**	3.52	1.29	4.81	7.84	4.30
	1000	1.67	1.68	4.37	**5.98**	5.94	3.71	1.42	5.90	8.01	5.00
	2000	1.78	1.91	5.36	7.93	**8.72**	4.06	1.20	5.43	10.5	6.32
Law school	250	10.9	22.2	16.4	**>100**	>100	31.8	10.7	>100	>100	>100
	500	12.8	19.7	37.6	**>100**	>100	52.4	9.46	>100	>100	>100
	1000	11.0	21.4	60.4	**>100**	>100	78.4	8.26	>100	>100	>100
	2000	10.7	20.4	**>100**	>100	>100	>100	0.00	>100	>100	>100
MAGIC	250	2.01	1.98	2.32	**3.93**	3.90	2.71	1.65	3.89	5.30	3.06
	500	2.11	2.26	2.76	**4.82**	4.57	3.81	1.71	4.89	5.94	3.27
	1000	2.01	2.10	3.16	**5.43**	5.33	4.69	1.61	4.73	6.75	4.60
EEG eye state	250	1.78	1.96	1.65	**3.55**	3.51	2.71	1.93	3.80	4.06	2.98
	500	1.94	2.00	1.83	**3.73**	3.44	3.33	1.82	4.23	3.99	3.15
	1000	2.02	2.16	1.93	**3.80**	3.61	3.68	1.91	4.06	4.86	3.04
Phoneme	250	2.13	2.29	2.77	4.60	**5.02**	2.98	1.77	4.68	6.74	3.19
	500	2.18	2.41	3.60	**5.79**	5.67	4.51	1.78	5.66	8.71	4.36
Spambase	250	2.29	2.61	5.55	**11.8**	11.3	5.56	2.19	16.1	20.0	5.69
Churn	250	1.56	1.49	5.99	**9.22**	8.41	2.96	1.42	5.77	10.3	9.48
	500	1.59	1.78	32.4	**60.8**	29.6	8.42	1.28	4.46	70.6	77.1
Electricity	250	1.83	1.84	2.65	**4.04**	3.83	2.76	1.63	4.88	5.26	2.97
	500	1.71	1.78	2.84	4.54	**4.66**	3.40	1.59	4.82	5.42	3.20
	1000	1.93	1.85	3.34	**4.97**	4.72	3.74	1.51	5.01	6.27	4.07
	2000	1.95	1.85	4.12	**5.85**	5.49	3.92	1.44	5.54	8.34	5.12

Table 5 Quality of inference by budget. Mean: coverage of nominal 95% intervals averaged over estimands and cells; worst: the lowest coverage of any estimand in any cell. "+ floor": Wald intervals with the zero-variance safeguard. Fail: largest share of samples with an undefined estimate (SRS only).

L	Cells	SRS		PATTERN		PATTERN-DIF (plain)		PATTERN-DIF (resid)		DIF (plain) + floor		DIF (resid) + floor		PATTERN + floor		Fail
		mean	worst	mean	worst	mean	worst	mean	worst	mean	worst	mean	worst	mean	worst	(max)
250	11	0.911	0.10	0.938	0.53	0.940	0.59	0.943	0.69	0.946	0.82	0.948	0.85	0.963	0.90	0.005
500	10	0.923	0.28	0.944	0.82	0.946	0.84	0.944	0.85	0.948	0.84	0.945	0.85	0.959	0.89	0.000
1000	8	0.938	0.71	0.950	0.89	0.952	0.90	0.949	0.90	0.952	0.90	0.950	0.90	0.959	0.91	0.000
2000	6	0.942	0.53	0.956	0.89	0.950	0.89	0.952	0.90	0.950	0.89	0.952	0.90	0.957	0.89	0.000

Table 6 Efficiency of PATTERN-DIF against SRS (ratio of mean squared errors) on four train/pool splits. Split 0 is the original one. Mean, standard deviation and range are over the four splits. The last rows give the median over the six settings. Source: `results/label_efficiency_splits.json`.

Setting	L	split 0	split 1	split 2	split 3	Mean	SD	Range
Adult (sex)	500	6.8	7.8	6.8	8.1	7.4	0.7	6.8–8.1
	1000	7.8	7.3	7.2	7.8	7.5	0.3	7.2–7.8
Adult (race)	500	12.4	10.6	12.8	14.6	12.6	1.6	10.6–14.6
	1000	14.7	14.9	16.4	15.6	15.4	0.8	14.7–16.4
Bank marketing	500	12.1	11.6	12.4	11.6	11.9	0.4	11.6–12.4
	1000	15.0	15.2	13.2	12.7	14.0	1.2	12.7–15.2
Credit card	500	5.3	5.6	6.1	6.7	5.9	0.6	5.3–6.7
	1000	6.0	6.5	7.9	6.7	6.8	0.8	6.0–7.9
MAGIC	500	4.8	4.5	5.4	5.0	4.9	0.4	4.5–5.4
	1000	5.4	5.2	5.2	5.8	5.4	0.3	5.2–5.8
Electricity	500	4.5	4.3	4.2	4.3	4.3	0.1	4.2–4.5
	1000	5.0	5.3	4.5	4.9	4.9	0.3	4.5–5.3
Median over settings	500	6.0	6.7	6.5	7.4			
Median over settings	1000	6.9	6.9	7.6	7.3			

Table 7 Coverage of the population contrast (nominal 0.95) over 108 contrasts, $R = 300$ pools per row. $V_{\text{pool}}/V_{\text{des}}$ is the median over contrasts of the estimated ratio. The last column is the coverage of the pool value by the pool-conditional interval. Source: `results/two_phase_check.json`.

			Pool-conditional		Two-phase		
N	L	$V_{\text{pool}}/V_{\text{des}}$	mean	worst	mean	worst	Pool value (pool-cond.) mean
2,000	250	0.50	0.899	0.86	0.949	0.92	0.949
	500	1.51	0.809	0.72	0.952	0.92	0.948
8,000	250	0.11	0.940	0.91	0.951	0.92	0.952
	500	0.25	0.921	0.87	0.947	0.90	0.948
	1000	0.58	0.889	0.83	0.950	0.92	0.945
	2000	1.57	0.792	0.64	0.949	0.92	0.953
20,000	250	0.04	0.940	0.90	0.945	0.91	0.945
	500	0.09	0.942	0.90	0.952	0.91	0.953
	1000	0.20	0.928	0.89	0.947	0.92	0.949
	2000	0.44	0.903	0.83	0.949	0.91	0.949

Table 8 Baselines by label budget L (number of settings in brackets). Efficiency: ratio of mean squared errors, median over the non-degenerate comparison estimands within a setting, then median over settings; "vs SRS" is MSE(SRS)/MSE(method), "vs PATTERN" is MSE(PATTERN)/MSE(method), so values above 1 favour the method. Coverage: nominal 95% Wald intervals, mean over estimands and settings; "worst": lowest single-estimand coverage over all settings and budgets. $R = 200$ replications per setting and budget. Source: `results/label_baselines.json`.

	$L = 250$ (11)		$L = 500$ (10)		$L = 1000$ (8)		$L = 2000$ (6)		All (s, L)		
Method	vs SRS	vs PAT.	vs SRS	vs PAT.	vs SRS	vs PAT.	vs SRS	vs PAT.	Cover.	Worst	ms
SRS	1	0.27	1	0.25	1	0.23	1	0.14	0.925	0.16	8
PATTERN	3.75	1	3.95	1	4.35	1	7.43	1	0.947	0.52	8
IS (B1)	3.09	0.79	2.77	0.74	3.48	0.67	4.30	0.54	0.950	0.60	9
Score-10 (B2)	1.45	0.40	2.09	0.52	2.24	0.56	4.16	0.50	0.941	0.17	8
Score-5 (B2)	1.97	0.54	2.24	0.57	2.15	0.52	3.18	0.43	0.943	0.37	8
Dis-25 (B3)	2.25	0.42	2.27	0.55	2.04	0.35	2.48	0.26	0.942	0.45	9
Dis-50 (B3)	1.84	0.43	1.94	0.37	1.68	0.34	1.93	0.25	0.940	0.36	9
PPI (B4)	1.82	0.49	1.82	0.47	1.86	0.44	1.81	0.27	0.940	0.27	34
PPI++ (B4)	2.10	0.49	1.97	0.48	1.93	0.44	1.89	0.27	0.929	0.16	49
StratPPI (B4)	3.07	1.01	4.45	1.12	4.58	1.08	8.50	0.83	0.924	0.14	44
Extension (B5): PATTERN sample, model-assisted estimator											
PATTERN-DIF (B5)	6.57	1.63	6.23	1.48	6.64	1.48	15.5	1.32	0.946	0.64	35
PATTERN-PPI (B5)	6.71	1.68	6.37	1.55	6.83	1.54	15.5	1.33	0.929	0.20	51

Table 9 Stratifier ablation (A1). Efficiency vs SRS (plain estimator): median over cells of the median over targets of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$. Rel.: median over the 29 cells of $\mathrm{MSE}_{\text{PATTERN-DIF}}/\mathrm{MSE}_{\text{variant}}$; Win: share of cells with Rel. > 1. Coverage: mean over cells of the mean over targets (mean) and of the worst target (worst), nominal 0.95. Source: `results/label_ablations_dif.json`.

	Efficiency vs SRS					Coverage	
Variant (estimator DIF unless stated)	$L = 250$	500	1000	Rel.	Win	mean	worst
PATTERN-DIF (default)	6.27	6.18	6.37	1.00	–	0.945	0.877
A1: stratifier							
Group × agree/disagree	4.33	4.65	4.81	0.74	0.07	0.946	0.881
Group × pattern of 3 models	2.82	2.98	3.43	0.38	0.00	0.941	0.867
Group $\times K\times$ 2 proxy bins	6.75	6.42	7.18	1.01	0.52	0.943	0.877
Group $\times K\times$ 3 proxy bins	6.79	5.90	6.61	0.99	0.38	0.945	0.890
PATTERN, residual allocation	5.64	6.02	6.68	0.96	0.45	0.945	0.886
References							
GROUP-DIF (strata = group)	2.10	2.04	2.07	0.28	0.00	0.941	0.850
SRS-DIF (PPI, $\lambda = 1$)	1.83	1.91	1.92	0.28	0.00	0.938	0.844
Proportional allocation (no proxy)	3.85	3.75	3.50	0.57	0.03	0.943	0.877
Gap-targeted allocation (A5)	5.36	5.26	5.22	0.74	0.00	0.941	0.866
PATTERN, plug-in (HT)	3.63	3.93	4.34	0.65	0.00	0.944	0.858

Table 10 Small-sample inference for PATTERN-DIF (A4). Mean: mean over cells and targets; worst: mean over cells of the worst-target coverage; min: minimum over cells of the worst-target coverage. Small pools: Churn, Spambase, Phoneme, Law school. Width: average interval width relative to Wald over all 29 cells. Floors: 'zero' replaces a zero stratum variance by the proxy residual variance, 'max' uses the larger of the two in every stratum.

	$L = 250$, all			$L = 250$, small		$L \geq 500$		
Method	mean	worst	min	worst	min	worst	min	Width
Wald (default)	0.941	0.852	0.633	0.827	0.633	0.892	0.833	1.00
Wald + floor 'zero'	0.947	0.876	0.780	0.893	0.873	0.893	0.833	1.00
Wald + floor 'max'	0.985	0.948	0.920	0.953	0.933	0.939	0.920	1.67
t, Satterthwaite df	0.945	0.865	0.700	0.850	0.700	0.893	0.833	1.01
t + floor 'zero'	0.950	0.885	0.787	0.905	0.887	0.894	0.833	1.01
Stratified bootstrap	0.950	0.878	0.767	0.893	0.767	0.918	0.860	1.11
References								
SRS-DIF (PPI), Wald	0.934	0.818	0.267	0.725	0.267	0.860	0.400	1.00
SRS-DIF, Wald + floor 'max'	0.976	0.929	0.900	0.942	0.900	0.929	0.880	1.23
SRS-DIF, bootstrap	0.924	0.741	0.100	0.678	0.100	0.818	0.153	1.12
PATTERN plug-in (HT), Wald	0.940	0.807	0.473	0.728	0.473	0.890	0.800	–